Tensor Decompositions

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1 Tensors

Recall the definition of a tensor. For our purposes, we will not need the full machinery of tensor product spaces, and it will be more useful to think of tensors concretely, as arrays of numbers.

Definition 1.1 (Tensor)

An N**-way tensor**, or **tensor of order** N, is a multidimensional array

$$\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N} \cong \mathbb{R}^{I_1} \otimes \mathbb{R}^{I_2} \otimes \cdots \otimes \mathbb{R}^{I_N} \tag{1}$$

with entries $x_{i_1 i_2 \cdots i_N} \in \mathbb{R}$ for $1 \leq i_n \leq I_n$. Each index is called a **mode** (or way), and I_n is the **dimension** of mode n.[a]

[a] In some fields, the order of the tensor is called its "rank." We will not use this convention, since the word rank will mean something quite different in the next section.

Since we are working in $\mathbb{R}^n$ with the standard inner product, we will *not* distinguish between $\mathbb{R}^n$ and its dual $(\mathbb{R}^n)^*$. That is, we won't care about covariant versus contravariant indices, and we will freely treat elements of $\mathbb{R}^n$ as column vectors or row vectors, whichever is more convenient. A vector is then a 1-way tensor, and a matrix is a 2-way tensor. Tensors of order 3 or higher are called *higher-order* tensors.

Example 1.1 (A 3-Tensor)

A 3-way tensor $\mathcal{X} \in \mathbb{R}^{3 \times 4 \times 2}$ can be thought of as a box of numbers, and the easiest way to write it down is to list its two 3×4 **frontal slices** $X_k \coloneqq (x_{ijk})_{i,j}$.

$$X_1 = \begin{bmatrix} 1 & 4 & 7 & 10 \\ 2 & 5 & 8 & 11 \\ 3 & 6 & 9 & 12 \end{bmatrix}, \qquad X_2 = \begin{bmatrix} 13 & 16 & 19 & 22 \\ 14 & 17 & 20 & 23 \\ 15 & 18 & 21 & 24 \end{bmatrix} \tag{2}$$

For instance, $x_{111} = 1$, $x_{231} = 8$, and $x_{342} = 24$. We will use this as our running example.

We will often work with tensors that have some extra structure in their indices, so let's define them now. The first requirement is that all of the modes have the same size.

Definition 1.2 (Cubical Tensor)

A tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ is **cubical** if every mode has the same size, i.e. $\mathcal{X} \in \mathbb{R}^{I \times I \times \cdots \times I}$.

Now we can say what it means for the entries to be symmetric. In general we only ask for this in some of the modes, which is only possible if those modes have the same size.

Definition 1.3 (Symmetric Tensor)

A tensor $\mathcal{X}$ is **symmetric** in a set of modes if its entries are invariant under any permutation of the indices in those modes. For example, $\mathcal{X} \in \mathbb{R}^{I \times I \times K}$ is symmetric in modes one and two if all of its frontal slices are symmetric,

$$X_k = X_k^T, \quad k = 1, \ldots, K \tag{3}$$

If it is symmetric in every mode, then we have the following.

Definition 1.4 (Supersymmetric Tensor)

A cubical tensor is **supersymmetric**[a] if its entries are invariant under any permutation of the indices. For example, $\mathcal{X} \in \mathbb{R}^{I \times I \times I}$ is supersymmetric if

$$x_{ijk} = x_{ikj} = x_{jik} = x_{jki} = x_{kij} = x_{kji}, \quad \forall i, j, k \tag{4}$$

[a] Some authors prefer to just call this symmetric.

Definition 1.5 (Diagonal Tensor)

A tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ is **diagonal** if $x_{i_1 i_2 \cdots i_N} \neq 0$ only if $i_1 = i_2 = \cdots = i_N$.

The entries $x_{ii\cdots i}$ of a cubical tensor make up its **superdiagonal**. An example is the cubical tensor with ones along the superdiagonal, which is diagonal and supersymmetric. For $N = 2$, these are the usual symmetric and diagonal matrices. Supersymmetric tensors are in bijection with homogeneous polynomials, much like symmetric matrices are in bijection with quadratic forms.

Now we want to multiply tensors. There are many ways of doing this, but we will only consider multiplying a tensor by a matrix (or vector) along a single mode.

Definition 1.6 (n-Mode Product)

Let $\mathcal{X} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$.

1. The **n-mode (matrix) product** of $\mathcal{X}$ with a matrix $U \in \mathbb{R}^{J \times I_n}$ is the tensor $\mathcal{X} \times_n U \in \mathbb{R}^{I_1 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_N}$ with entries

$$(\mathcal{X} \times_n U)_{i_1 \cdots i_{n-1}\, j\, i_{n+1} \cdots i_N} = \sum_{i_n=1}^{I_n} x_{i_1 i_2 \cdots i_N}\, u_{j i_n} \tag{5}$$

2. The **n-mode (vector) product** of $\mathcal{X}$ with a vector $v \in \mathbb{R}^{I_n}$ is the tensor $\mathcal{X} \,\bar{\times}_n\, v \in \mathbb{R}^{I_1 \times \cdots \times I_{n-1} \times I_{n+1} \times \cdots \times I_N}$ of order $N - 1$ with entries

$$(\mathcal{X} \,\bar{\times}_n\, v)_{i_1 \cdots i_{n-1}\, i_{n+1} \cdots i_N} = \sum_{i_n=1}^{I_n} x_{i_1 i_2 \cdots i_N}\, v_{i_n} \tag{6}$$

In the matrix case, each mode-n fiber of $\mathcal{X}$ gets multiplied by U, and in the vector case, we take the inner product of each mode-n fiber with v. When $\mathcal{X}$ defines a multilinear operator, the n-mode product is just a change of basis in mode n.

Now, is this something we've already seen? Yes. Since we are not distinguishing between $\mathbb{R}^n$ and its dual, summing over a pair of shared indices is exactly a **tensor contraction**. The n-mode product $\mathcal{X} \times_n U$ is the contraction of $\mathcal{X}$ and U along the index i_n of $\mathcal{X}$ and the column index of U, followed by moving the leftover row index j of U into position n. Similarly, $\mathcal{X} \,\bar{\times}_n\, v$ is the contraction of $\mathcal{X}$ with v along mode n, which is why the order drops by one.

Example 1.2 (Mode Products of the Running Tensor)

Let $\mathcal{X}$ be as in Example 1.1.

1. Let $U = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$. Then $\mathcal{Y} = \mathcal{X} \times_1 U \in \mathbb{R}^{2\times 4\times 2}$ has frontal slices $Y_k = UX_k$,

$$Y_1 = \begin{bmatrix} 22 & 49 & 76 & 103 \\ 28 & 64 & 100 & 136 \end{bmatrix}, \qquad Y_2 = \begin{bmatrix} 130 & 157 & 184 & 211 \\ 172 & 208 & 244 & 280 \end{bmatrix} \tag{7}$$

2. Let $v = (1, 2, 3, 4)^T$. Then

$$\mathcal{X} \bar{\times}_2 v = \begin{bmatrix} 70 & 190 \\ 80 & 200 \\ 90 & 210 \end{bmatrix} \in \mathbb{R}^{3\times 2} \tag{8}$$

For instance, the top-left entry is $\langle x_{1:1}, v\rangle = 1 + 8 + 21 + 40 = 70$.

1.1 Matricization

It is often useful to turn a tensor into a matrix so that we can use the linear algebra that we already know. This process is called **matricization**, **unfolding**, or **flattening**. In general, this can be any reordering of the entries of an N-way array into a matrix (e.g. a $2 \times 3 \times 4$ tensor can be arranged as a 6×4 or a 3×8 matrix), but we will only need a special case.

A **mode-n fiber** of $\mathcal{X}$ is a vector obtained by fixing every index except the nth, which is the higher-order analogue of the rows and columns of a matrix. For a 3-way tensor, the mode-1, mode-2, mode-3 fibers are the columns $x_{:jk}$, the rows $x_{i:k}$, and the tubes $x_{ij:}$, respectively.

Definition 1.7 (Mode-n Matricization)

The **mode-n matricization** of $\mathcal{X} \in \mathbb{R}^{I_1\times\cdots\times I_N}$ is the matrix $X_{(n)} \in \mathbb{R}^{I_n \times (I_1\cdots I_{n-1}I_{n+1}\cdots I_N)}$ whose columns are the mode-n fibers of $\mathcal{X}$. Precisely, the entry $x_{i_1 i_2 \cdots i_N}$ is mapped to the matrix entry (i_n, j) where

$$j = 1 + \sum_{\substack{k=1 \\ k\neq n}}^{N} (i_k - 1)J_k, \qquad J_k = \prod_{\substack{m=1 \\ m\neq n}}^{k-1} I_m \tag{9}$$

The formal notation is clunky, so let's just do an example.

Example 1.3 (Unfoldings of a $3 \times 4 \times 2$ Tensor)

Let $\mathcal{X} \in \mathbb{R}^{3\times 4\times 2}$ be the tensor from Example 1.1. Then the three mode-n unfoldings are

$$X_{(1)} = \begin{bmatrix} 1 & 4 & 7 & 10 & 13 & 16 & 19 & 22 \\ 2 & 5 & 8 & 11 & 14 & 17 & 20 & 23 \\ 3 & 6 & 9 & 12 & 15 & 18 & 21 & 24 \end{bmatrix} \in \mathbb{R}^{3\times 8} \tag{10}$$

$$X_{(2)} = \begin{bmatrix} 1 & 2 & 3 & 13 & 14 & 15 \\ 4 & 5 & 6 & 16 & 17 & 18 \\ 7 & 8 & 9 & 19 & 20 & 21 \\ 10 & 11 & 12 & 22 & 23 & 24 \end{bmatrix} \in \mathbb{R}^{4\times 6} \tag{11}$$

$$X_{(3)} = \begin{bmatrix} 1 & 2 & 3 & 4 & \cdots & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 & \cdots & 22 & 23 & 24 \end{bmatrix} \in \mathbb{R}^{2\times 12} \tag{12}$$

So $X_{(1)} = [X_1 \;\; X_2]$ is just the frontal slices placed side by side.

Different papers order the columns differently, but the specific permutation of the columns is not important

as long as we are consistent. Similarly, we can **vectorize** a tensor into $\text{vec}(\mathcal{X}) \in \mathbb{R}^{I_1 \cdots I_N}$, and in the example above $\text{vec}(\mathcal{X}) = (1, 2, \ldots, 24)^T$.

The n-mode product has a particularly nice description in terms of unfoldings. Each mode-n fiber gets multiplied by U, which is exactly what left multiplication does to the columns of $X_{(n)}$, so

$$\mathcal{Y} = \mathcal{X} \times_n U \iff Y_{(n)} = UX_{(n)} \tag{13}$$

Unfoldings are how we reduce statements about tensors to statements about matrices, and for this we need to know how several matrix products interact with them.

Several matrix products show up when relating tensors to their unfoldings, so let us define them.

Definition 1.8 (Kronecker, Khatri-Rao, Hadamard Products)

1. The **Kronecker product** of $A \in \mathbb{R}^{I \times J}$ and $B \in \mathbb{R}^{K \times L}$ is the $(IK) \times (JL)$ matrix

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1J}B \\ a_{21}B & a_{22}B & \cdots & a_{2J}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{I1}B & a_{I2}B & \cdots & a_{IJ}B \end{bmatrix} = \begin{bmatrix} a_1 \otimes b_1 & a_1 \otimes b_2 & \cdots & a_J \otimes b_L \end{bmatrix} \tag{14}$$

2. The **Khatri-Rao product** is the "matching columnwise" Kronecker product. For $A \in \mathbb{R}^{I \times K}$ and $B \in \mathbb{R}^{J \times K}$, it is the $(IJ) \times K$ matrix

$$A \odot B = \begin{bmatrix} a_1 \otimes b_1 & a_2 \otimes b_2 & \cdots & a_K \otimes b_K \end{bmatrix} \tag{15}$$

3. The **Hadamard product** is the elementwise product. For $A, B \in \mathbb{R}^{I \times J}$, it is the $I \times J$ matrix $(A * B)_{ij} = a_{ij} b_{ij}$.

If a, b are vectors, then the Khatri-Rao and Kronecker products coincide, $a \odot b = a \otimes b$. These products, along with the n-mode product, satisfy the following properties, which we will use over and over.

Lemma 1.1 (Properties of Tensor and Matrix Products)

Let $A^\dagger$ denote the Moore-Penrose pseudoinverse of A.

1. *(Kronecker)* For matrices of compatible sizes,

$$(A \otimes B)(C \otimes D) = AC \otimes BD \tag{16}$$

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger \tag{17}$$

2. *(Khatri-Rao)* The Khatri-Rao product is associative,

$$A \odot B \odot C = (A \odot B) \odot C = A \odot (B \odot C) \tag{18}$$

and,

$$(A \odot B)^T (A \odot B) = A^T A * B^T B \tag{19}$$

$$(A \odot B)^\dagger = \left((A^T A) * (B^T B)\right)^\dagger (A \odot B)^T \tag{20}$$

3. *(n-mode, distinct modes)* The order of multiplication in distinct modes is irrelevant,

$$\mathcal{X} \times_m A \times_n B = \mathcal{X} \times_n B \times_m A \qquad (m \neq n) \tag{21}$$

4. *(n-mode, same mode)*

$$\mathcal{X} \times_n A \times_n B = \mathcal{X} \times_n (BA) \tag{22}$$

5. *(n-mode, vectors)* For vectors, precedence matters since the modes get renumbered after each product. For $m < n$,

$$\mathcal{X} \bar{\times}_m a \bar{\times}_n b = (\mathcal{X} \bar{\times}_m a) \bar{\times}_{n-1} b = (\mathcal{X} \bar{\times}_n b) \bar{\times}_m a \tag{23}$$

6. *(Unfolding)* Let $\mathcal{X} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ and $A^{(n)} \in \mathbb{R}^{J_n \times I_n}$ for all n. Then for any n,

$$\mathcal{Y} = \mathcal{X} \times_1 A^{(1)} \times_2 A^{(2)} \cdots \times_N A^{(N)} \iff Y_{(n)} = A^{(n)} X_{(n)} \big(A^{(N)} \otimes \cdots \otimes A^{(n+1)} \otimes A^{(n-1)} \otimes \cdots \otimes A^{(1)}\big)^T \tag{24}$$

Proof. The Kronecker and Khatri-Rao identities are direct computations. For (3) and (4), note that by the unfolding relation $Y_{(n)} = UX_{(n)}$, multiplying by A and then B in mode n gives $B(AX_{(n)}) = (BA)X_{(n)}$, which is (4). Distinct modes act on different indices, so the sums can be exchanged, which is (3). The proof of (6) can be found in Kolda's paper on multilinear operators; TBD.

1.2 Tensor Rank

The basic building block of a tensor is the following.

Definition 1.9 (Outer Product)

The **outer product** of vectors $a^{(1)} \in \mathbb{R}^{I_1}, \ldots, a^{(N)} \in \mathbb{R}^{I_N}$ is the N-way tensor $a^{(1)} \circ a^{(2)} \circ \cdots \circ a^{(N)} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ with entries

$$\big(a^{(1)} \circ a^{(2)} \circ \cdots \circ a^{(N)}\big)_{i_1 i_2 \cdots i_N} = a^{(1)}_{i_1} a^{(2)}_{i_2} \cdots a^{(N)}_{i_N} \tag{25}$$

A tensor that can be written this way is called **rank-one**.

For $N = 2$, $a \circ b = ab^T$ is the usual outer product of vectors, which is exactly a rank-1 matrix. So we are just generalizing what we already know: every matrix of rank D can be written as a sum of D rank-one matrices, and the rank is the smallest such D. We take this to be the definition for tensors.

Definition 1.10 (Tensor Rank)

The **rank** of a tensor $\mathcal{X}$, denoted $\text{rank}(\mathcal{X})$, is the smallest D such that $\mathcal{X}$ can be written as a sum of D rank-one tensors,

$$\text{rank}(\mathcal{X}) \coloneqq \min \left\{ D : \mathcal{X} = \sum_{d=1}^{D} a^{(1)}_d \circ a^{(2)}_d \circ \cdots \circ a^{(N)}_d \right\} \tag{26}$$

This looks like a perfect analogue of matrix rank, but the two behave very differently. For a matrix, we can compute the rank with Gaussian elimination or the SVD in polynomial time. For tensors, *there is no straightforward algorithm to determine the rank of a given tensor*; in fact, the problem is NP-hard. As a sign of how bad it is, there is a particular $9 \times 9 \times 9$ tensor whose rank is only known to lie between 18 and 23. In practice, the rank is determined numerically by trying to fit rank-D models for various D. Some other differences are the following.

1. *Rank depends on the field.* The rank of a real tensor can be different over $\mathbb{R}$ and $\mathbb{C}$, whereas for matrices it cannot.

Example 1.4 (Rank Over $\mathbb{R}$ vs. $\mathbb{C}$)

Let $\mathcal{X} \in \mathbb{R}^{2\times 2\times 2}$ have frontal slices

$$X_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad X_2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \tag{27}$$

Then $\mathcal{X}$ has rank 3 over $\mathbb{R}$ and rank 2 over $\mathbb{C}$. Writing $\mathcal{X} = \sum_d a_d \circ b_d \circ c_d$ and stacking the vectors into the columns of factor matrices A, B, C, a rank decomposition over $\mathbb{R}$ is

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} \tag{28}$$

whereas over $\mathbb{C}$ we can do it with only two components,

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix}, \quad B = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \tag{29}$$

The proof that the real rank is not 2 is more involved (this example is due to Kruskal), and we omit it.

2. *Maximum rank is not typical rank.* The **maximum rank** is the largest rank attainable, while a **typical rank** is any rank that occurs with positive probability (i.e. on a set of positive Lebesgue measure). For $I \times J$ matrices, both are equal to $\min\{I, J\}$. For tensors they can differ, and moreover over $\mathbb{R}$ there can be more than one typical rank (over $\mathbb{C}$ there is always exactly one).

Example 1.5 (Typical Ranks of $2 \times 2 \times 2$ Tensors)

If we draw every entry of a $2 \times 2 \times 2$ tensor iid from a standard normal, then Monte Carlo experiments show that about 79% of them have rank 2 and about 21% have rank 3. So both 2 and 3 are typical ranks over $\mathbb{R}$. Rank-one tensors are possible, but occur with probability zero.

3. *Weak upper bounds.* For a general $\mathcal{X} \in \mathbb{R}^{I\times J\times K}$, the only known bound on the maximum rank is the weak one
$$\operatorname{rank}(\mathcal{X}) \leq \min\{IJ, IK, JK\} \tag{30}$$
compared to the matrix case where it is simply $\min\{I, J\}$.

Another important difference, which concerns the uniqueness of the decomposition, will be discussed when we introduce the CP decomposition.

2 Decompositions

2.1 CP Decomposition

We can now state the first of the two main tensor decompositions. The idea is to write a tensor as a sum of rank-one tensors, and in this sense it generalizes the SVD, where $X = \sum_d \sigma_d u_d v_d^T$. It goes by many names: the polyadic form of a tensor (Hitchcock, 1927), PARAFAC for parallel factors (Harshman, 1970), CANDECOMP for canonical decomposition (Carroll and Chang, 1970), and now just **CP** for CANDECOMP/PARAFAC.

Definition 2.1 (CP Decomposition)

A **CP decomposition** of $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$ with D components is an approximation

$$\mathcal{X} \approx \sum_{d=1}^{D} a_d \circ b_d \circ c_d =: [\![A, B, C]\!] \tag{31}$$

where $a_d \in \mathbb{R}^I$, $b_d \in \mathbb{R}^J$, $c_d \in \mathbb{R}^K$. Elementwise, this is $x_{ijk} \approx \sum_{d=1}^{D} a_{id} b_{jd} c_{kd}$. The matrices $A = [a_1 \ \cdots \ a_D] \in \mathbb{R}^{I \times D}$, and likewise $B \in \mathbb{R}^{J \times D}$, $C \in \mathbb{R}^{K \times D}$, are called the **factor matrices**. More generally, for $\mathcal{X} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$,

$$\mathcal{X} \approx [\![\lambda; A^{(1)}, \ldots, A^{(N)}]\!] \coloneqq \sum_{d=1}^{D} \lambda_d \, a_d^{(1)} \circ a_d^{(2)} \circ \cdots \circ a_d^{(N)} \tag{32}$$

where $A^{(n)} \in \mathbb{R}^{I_n \times D}$ has columns normalized to unit length and the weights are absorbed into $\lambda \in \mathbb{R}^D$.

If the approximation is an equality and $D = \operatorname{rank}(\mathcal{X})$, then this is called a **rank decomposition**. Using the matricization and Khatri-Rao products above, the CP decomposition can be written in matrix form, one for each mode.

$$X_{(1)} \approx A(C \odot B)^T \tag{33}$$

$$X_{(2)} \approx B(C \odot A)^T \tag{34}$$

$$X_{(3)} \approx C(B \odot A)^T \tag{35}$$

and for general N, with $\Lambda = \operatorname{diag}(\lambda)$,

$$X_{(n)} \approx A^{(n)} \Lambda \big(A^{(N)} \odot \cdots \odot A^{(n+1)} \odot A^{(n-1)} \odot \cdots \odot A^{(1)}\big)^T \tag{36}$$

2.1.1 Uniqueness

Here is one more major difference between matrices and tensors, and this time it's in the tensors' favor. Matrix decompositions are not unique. If $X = AB^T$ has rank D, then for any orthogonal $W \in \mathbb{R}^{D \times D}$ we also have $X = (AW)(BW)^T$. The SVD is unique (when the singular values are distinct) only because we add orthogonality constraints and order the singular values. The CP decomposition, on the other hand, is unique under much weaker conditions, up to two trivial indeterminacies.

1. *Permutation.* The rank-one components can be reordered, $[\![A, B, C]\!] = [\![A\Pi, B\Pi, C\Pi]\!]$ for any permutation matrix Π.
2. *Scaling.* We can rescale each component, $\sum_d (\alpha_d a_d) \circ (\beta_d b_d) \circ (\gamma_d c_d)$, as long as $\alpha_d \beta_d \gamma_d = 1$ for all d.

Theorem 2.1 (Kruskal)

Let $\mathcal{X} = [\![A, B, C]\!]$ with D components. Let the k**-rank** k_A of a matrix A be the largest k such that any k columns of A are linearly independent. If

$$k_A + k_B + k_C \geq 2D + 2 \tag{37}$$

then the decomposition is unique up to permutation and scaling.

2.2 Tucker Decomposition

The second decomposition is a higher-order version of PCA. Instead of a sum of rank-one tensors, it writes the tensor as a small *core* tensor that is transformed by a matrix along each mode. It is also called three-mode factor analysis, three-mode PCA, N-mode PCA, or higher-order SVD (HOSVD).

Definition 2.2 (Tucker Decomposition)

A **Tucker decomposition** of $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$ is an approximation

$$\mathcal{X} \approx \mathcal{G} \times_1 A \times_2 B \times_3 C = \sum_{p=1}^{R_1} \sum_{q=1}^{R_2} \sum_{r=1}^{R_3} g_{pqr}\, a_p \circ b_q \circ c_r =: [\![\mathcal{G}; A, B, C]\!] \tag{38}$$

where $\mathcal{G} \in \mathbb{R}^{R_1 \times R_2 \times R_3}$ is the **core tensor**, and $A \in \mathbb{R}^{I \times R_1}$, $B \in \mathbb{R}^{J \times R_2}$, $C \in \mathbb{R}^{K \times R_3}$ are the **factor matrices**, which are usually taken to have orthonormal columns. For $\mathcal{X} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$,

$$\mathcal{X} \approx \mathcal{G} \times_1 A^{(1)} \times_2 A^{(2)} \cdots \times_N A^{(N)} = [\![\mathcal{G}; A^{(1)}, \ldots, A^{(N)}]\!] \tag{39}$$

with $\mathcal{G} \in \mathbb{R}^{R_1 \times \cdots \times R_N}$ and $A^{(n)} \in \mathbb{R}^{I_n \times R_n}$.

The columns of the factor matrices can be thought of as principal components in each mode, and the entries of the core $\mathcal{G}$ describe the level of interaction between the components of different modes. If $R_n < I_n$, then $\mathcal{G}$ is a compressed version of $\mathcal{X}$, and the storage needed for $(\mathcal{G}, A^{(1)}, \ldots, A^{(N)})$ can be much smaller than for $\mathcal{X}$. By the Unfolding property in Lemma 1.1, the matricized forms are

$$X_{(n)} \approx A^{(n)} G_{(n)} \left(A^{(N)} \otimes \cdots \otimes A^{(n+1)} \otimes A^{(n-1)} \otimes \cdots \otimes A^{(1)}\right)^T \tag{40}$$

Lemma 2.2 (CP is a Special Case of Tucker)

The CP decomposition $[\![A, B, C]\!]$ with D components is the Tucker decomposition with $R_1 = R_2 = R_3 = D$ and a diagonal core $\mathcal{G}$ with ones on the superdiagonal.

Proof. With $g_{pqr} = 1$ if $p = q = r$ and 0 otherwise, the triple sum collapses to $\sum_d a_d \circ b_d \circ c_d$.

Just as the rank of a matrix is the dimension of its column space, we can define the following for Tucker.

Definition 2.3 (n-Rank)

The **n-rank** of $\mathcal{X}$, denoted $\text{rank}_n(\mathcal{X})$, is the column rank of $X_{(n)}$, i.e. the dimension of the span of the mode-n fibers. If $R_n = \text{rank}_n(\mathcal{X})$, then $\mathcal{X}$ is a **rank-$(R_1, \ldots, R_N)$** tensor.

Trivially $R_n \leq I_n$. Note that this is *not* the same as the tensor rank defined earlier. It is always easy to find an exact Tucker decomposition of $\mathcal{X}$ with $R_n = \text{rank}_n(\mathcal{X})$. If we pick some $R_n < \text{rank}_n(\mathcal{X})$, then the decomposition is necessarily inexact (a *truncated* Tucker decomposition) and harder to compute.

Finally, we can restrict some of the factor matrices to be the identity. The **Tucker2** decomposition sets one factor matrix to the identity, e.g. $\mathcal{X} \approx \mathcal{G} \times_1 A \times_2 B = [\![\mathcal{G}; A, B, I]\!]$, and **Tucker1** sets two of them, e.g. $\mathcal{X} \approx \mathcal{G} \times_1 A$, which has $X_{(1)} \approx A G_{(1)}$ and is therefore equivalent to ordinary PCA on the mode-1 unfolding.

3 Rank of a Given Tensor

We have seen that, unlike matrices, there is no straightforward algorithm for computing $\text{rank}(\mathcal{X})$. Before talking about methods, let's see what it takes to find the rank of one small tensor by hand.

Example 3.1 (Rank of the Running Tensor)

Let $\mathcal{X} \in \mathbb{R}^{3\times4\times2}$ be the tensor from Example 1.1, with frontal slices

$$X_1 = \begin{bmatrix} 1 & 4 & 7 & 10 \\ 2 & 5 & 8 & 11 \\ 3 & 6 & 9 & 12 \end{bmatrix}, \qquad X_2 = \begin{bmatrix} 13 & 16 & 19 & 22 \\ 14 & 17 & 20 & 23 \\ 15 & 18 & 21 & 24 \end{bmatrix} \tag{41}$$

Note that $X_2 = X_1 + 12\,\mathbf{1}_3\mathbf{1}_4^T$. The entries of X_1 are $x_{ij1} = i + 3(j-1)$, so

$$X_1 = p\mathbf{1}_4^T + \mathbf{1}_3 q^T = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 3 & 6 & 9 \\ 0 & 3 & 6 & 9 \\ 0 & 3 & 6 & 9 \end{bmatrix} \tag{42}$$

where $p = (1,2,3)^T$, $q = (0,3,6,9)^T$, and $\mathbf{1}_n$ is the all-ones vector. Then we have the following bounds.

1. *Upper bound.* Since X_1 is the sum of two rank-one matrices, and $X_2 = X_1 + 12\,\mathbf{1}_3\mathbf{1}_4^T$ adds one more, we can write

$$\mathcal{X} = p \circ \mathbf{1}_4 \circ (1,1) + \mathbf{1}_3 \circ q \circ (1,1) + 12\,\mathbf{1}_3 \circ \mathbf{1}_4 \circ (0,1) \tag{43}$$

   so $\text{rank}(\mathcal{X}) \le 3$.

2. *Lower bound.* Suppose the rank were 2, so $\mathcal{X} = a_1 \circ b_1 \circ c_1 + a_2 \circ b_2 \circ c_2$. Then every slice is a combination of the same two rank-one matrices,

$$X_k = c_{k1}\, a_1 b_1^T + c_{k2}\, a_2 b_2^T, \qquad k = 1,2 \tag{44}$$

   Since X_1, X_2 are linearly independent, the span of X_1, X_2 would then be *exactly* the span of two rank-one matrices, so it would contain at least two non-parallel rank-one matrices. But every matrix in the span of X_1, X_2 has the form $uX_1 + v\,\mathbf{1}_3\mathbf{1}_4^T$, which equals $u\,p\mathbf{1}_4^T + \mathbf{1}_3(uq + v\mathbf{1}_4)^T$. If $u \neq 0$, this has rank 2, since $p, \mathbf{1}_3$ are independent and $\mathbf{1}_4, uq + v\mathbf{1}_4$ are independent. So the only rank-one matrices in the span are the multiples of $\mathbf{1}_3\mathbf{1}_4^T$, a contradiction. Hence $\text{rank}(\mathcal{X}) = 3$.

Notice what the lower bound argument did. We never really used the third mode, except through the fact that the slices are combinations of a few rank-one matrices. The argument in the example above is the key idea behind almost every method below. It turns the rank of a 3-tensor into a statement about a subspace of matrices.

Lemma 3.1 (Slice-Span Characterization)

Let $\mathcal{X} \in \mathbb{R}^{I\times J\times K}$ with frontal slices $X_1, \dots, X_K \in \mathbb{R}^{I\times J}$. Then $\text{rank}(\mathcal{X})$ is the smallest r such that there exist r rank-one matrices $M_1, \dots, M_r \in \mathbb{R}^{I\times J}$ with

$$\text{span}\{X_1, \dots, X_K\} \subseteq \text{span}\{M_1, \dots, M_r\} \tag{45}$$

Proof. The idea is to look at what a sum of r rank-one tensors does to a single slice. Suppose $\mathcal{X} = \sum_{d=1}^r a_d \circ b_d \circ c_d$, so that $x_{ijk} = \sum_d a_{id} b_{jd} c_{kd}$. Fixing k, the (i,j) entry of the kth slice is

$$(X_k)_{ij} = \sum_{d=1}^r c_{kd}\, a_{id} b_{jd} \implies X_k = \sum_{d=1}^r c_{kd}\, a_d b_d^T \tag{46}$$

That is, every slice is a linear combination of the *same* r rank-one matrices $M_d \coloneqq a_d b_d^T$, and the only thing that changes from slice to slice are the coefficients c_{kd}, i.e. the kth row of the third factor. In particular, all of the slices lie in $\operatorname{span}\{M_1, \dots, M_r\}$, which shows that the smallest such r is at most the rank. (In the example, the M_d were $p\mathbf{1}_4^T$, $\mathbf{1}_3 q^T$, and $\mathbf{1}_3\mathbf{1}_4^T$.)
Conversely, suppose we are given rank-one matrices $M_d = a_d b_d^T$ for $d = 1, \dots, r$ whose span contains every slice. Then each slice can be written in terms of them, $X_k = \sum_d c_{kd} M_d$, for some coefficients c_{kd}. We now just collect these coefficients into vectors $c_d = (c_{1d}, \dots, c_{Kd})^T \in \mathbb{R}^K$. By the computation above, the tensor $\sum_d a_d \circ b_d \circ c_d$ has the slices $\sum_d c_{kd} a_d b_d^T = X_k$, so it is equal to $\mathcal{X}$. This is a decomposition with r terms, so the rank is at most r.

In other words, the third factor c_d is not something we need to search for; it is just bookkeeping for how to express the slices in terms of the M_d's. So computing tensor rank is the problem of finding the fewest rank-one matrices whose span contains a given subspace of matrices. For $K = 1$ this is the matrix rank, since the span of a single matrix X_1 is contained in the span of r rank-one matrices exactly when $\operatorname{rank}(X_1) \le r$. For $K = 2$ it becomes tractable, as we will see.

3.1 Hardness

The first thing to understand is why there is no algorithm.

Theorem 3.2 (Håstad, 1990)

Determining whether a 3-tensor over $\mathbb{Q}$ has rank at most r is NP-hard. Over a finite field it is NP-complete.

Hillar and Lim (2013, *Most Tensor Problems Are NP-Hard*) observed that Håstad's argument also works over $\mathbb{R}$ and $\mathbb{C}$, and showed that many other tensor problems are NP-hard as well, e.g. best rank-one approximation, the spectral norm, and eigenvalues of tensors.

So we should lower our expectations. We cannot hope for an efficient algorithm that works on all tensors, so the goal becomes finding good bounds and special families where the rank *is* computable.

3.2 Lower Bounds

3.2.1 Flattening

The simplest lower bound is the one we already have access to.

Lemma 3.3 (Flattening Bound)

For any $\mathcal{X}$ and any mode n,

$$\operatorname{rank}(\mathcal{X}) \ge \operatorname{rank}\big(X_{(n)}\big) = \operatorname{rank}_n(\mathcal{X}) \tag{47}$$

Proof. If $\mathcal{X} = \sum_{d=1}^r a_d^{(1)} \circ \cdots \circ a_d^{(N)}$, then by the CP matricization formula $X_{(n)} = A^{(n)}\big(A^{(N)} \odot \cdots \odot A^{(n+1)} \odot A^{(n-1)} \odot \cdots \odot A^{(1)}\big)^T$, which is a product through an r-dimensional space and hence has rank at most r.

This bound is sharp for matrices but very weak for tensors. Let's see why.

Example 3.2 (Sharp for Matrices, Weak for Tensors)

1. *Matrices.* If $N = 2$, then $X_{(1)} = X$ and $X_{(2)} = X^T$, so the bound says $\text{rank}(X) \geq \text{rank}(X)$, which is an equality. In other words, the rank of a matrix *is* the rank of its flattenings, and it can be computed by Gaussian elimination. (This is just the statement that a rank factorization $X = AB^T$ with $\text{rank}(X)$ columns exists.)
2. *Tensors.* Let $\mathcal{X}$ be the running tensor of Example 3.1, which has rank 3. The fibers of $\mathcal{X}$ lie in small subspaces.
   - The mode-1 fibers (columns of X_1 and X_2) all lie in $\text{span}\{p, \mathbf{1}_3\}$, so $\text{rank}_1(\mathcal{X}) = 2$.
   - The mode-2 fibers (rows of X_1, X_2) all lie in $\text{span}\{\mathbf{1}_4, q\}$, so $\text{rank}_2(\mathcal{X}) = 2$.
   - The mode-3 fibers are $(a, a + 12)$, which span $\mathbb{R}^2$, so $\text{rank}_3(\mathcal{X}) = 2$.

   So all three flattenings have rank 2, and the flattening bound only certifies $\text{rank}(\mathcal{X}) \geq 2$, while the true rank is 3.

The weakness is not a quirk of this example. Since $\text{rank}_n(\mathcal{X}) \leq I_n$, the bound can never certify a rank larger than $\max\{I_1, \dots, I_N\}$, whereas the rank of an $N \times N \times N$ tensor can be on the order of N^2. For example, a generic $3 \times 3 \times 3$ tensor has all three flattenings of rank at most 3, but its rank is 5.

So far we only have bounds that don't meet. Let's look at the simplest families of tensors where a lower bound and an upper bound do meet, so that we can actually say what the rank is. The first is the easiest case of all.

Lemma 3.4 (Rank-One Tensors)

A nonzero tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ has rank one if and only if $\text{rank}_n(\mathcal{X}) = 1$ for every mode n, i.e. if and only if in every mode all of the fibers are multiples of a single vector.

Proof. If $\mathcal{X} = a^{(1)} \circ \cdots \circ a^{(N)}$ then every mode-n fiber is a multiple of $a^{(n)}$, so $\text{rank}_n(\mathcal{X}) = 1$. For the converse we use induction on N, where the case $N = 2$ is the statement that a matrix of rank one is ab^T. Suppose $\text{rank}_1(\mathcal{X}) = 1$. Then all mode-1 fibers are multiples of one vector $a \neq 0$, which says that $\mathcal{X} = a \circ \mathcal{W}$ for some $\mathcal{W} \in \mathbb{R}^{I_2 \times \cdots \times I_N}$, i.e. $x_{i_1 i_2 \cdots i_N} = a_{i_1} w_{i_2 \cdots i_N}$. For $n \geq 2$, each mode-n fiber of $\mathcal{X}$ is a_{i_1} times a mode-n fiber of $\mathcal{W}$ and $a \neq 0$, so the fibers of $\mathcal{X}$ and $\mathcal{W}$ span the same space and $\text{rank}_n(\mathcal{W}) = \text{rank}_n(\mathcal{X}) = 1$. By induction $\mathcal{W}$ is rank one, so $\mathcal{X} = a \circ \mathcal{W}$ is too.

For example, the tensor with slices $X_1 = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $X_2 = \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix}$ has all fibers parallel to $(1, 2)$, $(1, 2)$, and $(1, 3)$ in modes $1, 2, 3$, respectively, and indeed $\mathcal{X} = (1, 2) \circ (1, 2) \circ (1, 3)$.

The next simplest family is the tensors that are diagonal, and here the flattening bound is exactly right.

Lemma 3.5 (Diagonal Tensors)

Let $\mathcal{X} \in \mathbb{R}^{I \times I \times \cdots \times I}$ be a diagonal tensor with r nonzero entries $x_{ii \cdots i} = \lambda_i$. Then $\text{rank}(\mathcal{X}) = r$.

Proof. Let S be the set of indices i with $\lambda_i \neq 0$, so $|S| = r$. For the upper bound,

$$\mathcal{X} = \sum_{i \in S} \lambda_i \, e_i \circ e_i \circ \cdots \circ e_i \tag{48}$$

is a sum of r rank-one tensors. For the lower bound, look at the flattening $X_{(1)}$. Its entry in row i and in the column indexed by $(i_2, \dots, i_N)$ is nonzero only if $i_2 = \cdots = i_N = i$. So for each $i \in S$ the column indexed by $(i, \dots, i)$ has a single nonzero entry λ_i, in row i, and all other columns are zero. These r columns are linearly independent, so $\text{rank}(X_{(1)}) = r$, and by the Flattening Bound, $\text{rank}(\mathcal{X}) \geq r$.

In particular the unit tensor $\sum_{i=1}^{n} e_i \circ e_i \circ e_i \in \mathbb{R}^{n\times n\times n}$, which is the tensor analogue of the identity matrix, has rank n, which is just as large as the flattening bound allows. So the Flattening Bound is not always weak. It is sharp for matrices and for diagonal tensors, and the running tensor above is an example where it fails.

Both of these are special cases of a simple principle, which is also easy to prove.

Lemma 3.6 (Contraction Does Not Increase Rank)

For any matrix U and any mode n, $\operatorname{rank}(\mathcal{X} \times_n U) \le \operatorname{rank}(\mathcal{X})$, with equality if U is invertible. Similarly, $\operatorname{rank}(\mathcal{X} \,\bar{\times}_n\, v) \le \operatorname{rank}(\mathcal{X})$ for any vector v.

Proof. If $\mathcal{X} = \sum_{d=1}^{r} a_d^{(1)} \circ \cdots \circ a_d^{(N)}$, then multiplying in mode n just multiplies the nth vector of each term by U,

$$\mathcal{X} \times_n U = \sum_{d=1}^{r} a_d^{(1)} \circ \cdots \circ \left(U a_d^{(n)}\right) \circ \cdots \circ a_d^{(N)} \tag{49}$$

which is a sum of r rank-one (or zero) tensors. For the vector case replace $U a_d^{(n)}$ by the scalar $\langle v, a_d^{(n)} \rangle$. If U is invertible, apply the inequality again with U^{-1}.

So the rank is unchanged by any change of basis in each mode, and the rank of a tensor is at least the rank of any subtensor obtained by contracting or selecting entries. For example, $\operatorname{rank}(a \circ Y) = \operatorname{rank}(Y)$ for $a \neq 0$ and a matrix Y, since contracting $a \circ Y$ with a vector u with $\langle a, u\rangle = 1$ gives back Y.

Returning to flattenings, more general flattenings, where we group modes in other ways, are no better than the min of the dimensions of the groups. Since the rank of a matrix is lower semicontinuous, the same bound holds for the border rank defined below.

Strassen's idea (1983) was to use flattenings that are not the standard ones. Landsberg and Ottaviani later systematized this with **Young flattenings**, of which the most used are the **Koszul flattenings**. For $\mathcal{T} \in A \otimes B \otimes C$ with $\dim A = 2p+1$, the map $\mathcal{T}_A^{\wedge p} : \Lambda^p A \otimes B^* \to \Lambda^{p+1} A \otimes C$ obtained by applying $\mathcal{T}$ and then wedging satisfies

$$\underline{\operatorname{rank}}(\mathcal{T}) \ge \frac{\operatorname{rank}\left(\mathcal{T}_A^{\wedge p}\right)}{\binom{2p}{p}} \tag{50}$$

The proof idea is that the flattening map is linear in $\mathcal{T}$, and for a rank-one tensor $a\otimes b\otimes c$ its rank is at most $\binom{2p}{p}$, so subadditivity of matrix rank gives the bound for a sum of r such tensors, and lower semicontinuity extends it to the border rank.

3.2.2 The Substitution Method

This method goes back to Pan (1966) and has been the main source of rank lower bounds in algebraic complexity (see Bürgisser–Clausen–Shokrollahi). The idea is to use one rank-one component of a decomposition to eliminate an entire slice.

Lemma 3.7 (Substitution)

Let $\mathcal{X} \in \mathbb{R}^{I\times J\times K}$ with horizontal slices $X^{(1)}, \dots, X^{(I)} \in \mathbb{R}^{J\times K}$, and suppose $X^{(1)} \neq 0$. Then

$$\operatorname{rank}(\mathcal{X}) \ge 1 + \min_{\lambda_2,\dots,\lambda_I \in \mathbb{R}} \operatorname{rank}\left(\mathcal{X}'_\lambda\right) \tag{51}$$

where $\mathcal{X}'_\lambda \in \mathbb{R}^{(I-1)\times J\times K}$ has slices $X^{(i)} + \lambda_i X^{(1)}$ for $i = 2, \dots, I$.

Proof. Take a decomposition $\mathcal{X} = \sum_{d=1}^{r} a_d \circ b_d \circ c_d$ with $r = \text{rank}(\mathcal{X})$. Then $X^{(i)} = \sum_d a_{id}\, b_d c_d^T$. Since $X^{(1)} \neq 0$, there is some d_0 with $a_{1d_0} \neq 0$. Setting $\lambda_i = -a_{id_0}/a_{1d_0}$, the slice $X^{(i)} + \lambda_i X^{(1)}$ has no contribution from component d_0, so $\mathcal{X}'_\lambda$ is a sum of the remaining $r-1$ rank-one tensors.

Note that we can choose *any* nonzero slice, in any mode, as the one to eliminate, and then iterate.

Example 3.3 (Rank of the W Tensor)

Let $\mathcal{W} = e_1 \circ e_1 \circ e_2 + e_1 \circ e_2 \circ e_1 + e_2 \circ e_1 \circ e_1 \in \mathbb{R}^{2\times 2\times 2}$, whose horizontal slices are

$$W^{(1)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad W^{(2)} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \tag{52}$$

If we eliminate with $W^{(2)}$, the remaining slice is $W^{(1)} + \lambda W^{(2)} = \begin{bmatrix} \lambda & 1 \\ 1 & 0 \end{bmatrix}$, which has determinant -1, hence rank 2 for every λ. By Lemma 3.7, $\text{rank}(\mathcal{W}) \geq 1 + 2 = 3$. (Eliminating with $W^{(1)}$ instead would only give 2, since $\lambda = 0$ gives rank one, so the choice of slice matters.) And clearly $\text{rank}(\mathcal{W}) \leq 3$, so it is exactly 3.

The same argument gives the rank of the diagonal (unit) tensor $\sum_{d=1}^{n} e_d \circ e_d \circ e_d$ as n. Border versions of the method (Landsberg–Michałek) combine it with Koszul flattenings and give the best known border rank lower bound for $n \times n$ matrix multiplication, $\underline{\text{rank}}(M_{\langle n,n,n\rangle}) \geq 2n^2 - \lceil \log_2 n \rceil - 1$.[1]

3.3 Exactly Solvable Case: Pencils

For $K = 2$ the problem is fully solved, and the answer is in terms of familiar matrix invariants. Here $\mathcal{X} \in \mathbb{R}^{I\times J\times 2}$ is just a *matrix pencil* (X_1, X_2), and its rank is determined by the Kronecker canonical form of the pencil $X_1 + tX_2$ (Grigoriev and JáJá, 1970s–80s; Ten Berge and Kiers extended to describe typical ranks). We prove the most important special case.

Theorem 3.8 (Rank of a Regular Pencil)

Let $\mathcal{X} \in \mathbb{F}^{I\times I\times 2}$ with X_1 invertible. Then $\text{rank}(\mathcal{X}) \geq I$, with equality if and only if $X_1^{-1}X_2$ is diagonalizable over $\mathbb{F}$.

Proof. The lower bound is the flattening bound since $\text{rank}(X_{(1)}) \geq \text{rank}(X_1) = I$.
($\Leftarrow$) Suppose $X_1^{-1}X_2 = P\Lambda P^{-1}$ with Λ diagonal. Let $A = X_1 P$ and $B = P^{-T}$. Then $AB^T = X_1$ and $A\Lambda B^T = X_1 P\Lambda P^{-1} = X_2$, so each slice is AD_kB^T with $D_1 = I, D_2 = \Lambda$ diagonal. That is a sum of I rank-one tensors.
($\Rightarrow$) If the rank is I, then by Lemma 3.1, $X_k = AD_kB^T$ for some $A, B \in \mathbb{F}^{I\times I}$ and diagonal D_k. Since X_1 is invertible, so are A, B, and D_1. Then $X_1^{-1}X_2 = B^{-T}D_1^{-1}D_2B^T$ is diagonalizable over $\mathbb{F}$.

Example 3.4 (Why Kruskal's Example Has Rank 3 Over $\mathbb{R}$)

In Example 1.4, $X_1 = I$ and $X_2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, so $X_1^{-1}X_2 = X_2$ has eigenvalues $\pm i$. It is diagonalizable over $\mathbb{C}$ but not over $\mathbb{R}$, so by Theorem 3.8, the rank is 2 over $\mathbb{C}$ and strictly more than 2 over $\mathbb{R}$. The

[1] `https://par.nsf.gov/servlets/purl/10094859`. For $n = 3$ this gives 15, improving the lower bound of 14 cited in the review.

explicit decomposition given earlier shows that it is exactly 3.

This also explains the large literature on typical ranks of $I \times J \times 2$ tensors: whether the random matrix $X_1^{-1}X_2$ has real eigenvalues is an event of positive probability, which is why there are two typical ranks over $\mathbb{R}$.

3.4 Upper Bounds and Algorithms

3.4.1 Simultaneous Diagonalization

The generalization of Theorem 3.8 to $K \geq 2$ is an actual algorithm, often called **Jennrich's algorithm**.

Theorem 3.9 (Simultaneous Diagonalization)

Let $\mathcal{X} = [\![A, B, C]\!]$ with $A \in \mathbb{R}^{I \times R}$ and $B \in \mathbb{R}^{J \times R}$ of full column rank R and any two columns of $C \in \mathbb{R}^{K \times R}$ linearly independent (i.e. $k_C \geq 2$). Then $\operatorname{rank}(\mathcal{X}) = R$ and the decomposition is unique up to permutation and scaling, and it can be computed with two eigendecompositions.

Proof. Pick random $u, v \in \mathbb{R}^K$ and contract the third mode,

$$M_u := \mathcal{X} \bar{\times}_3 u = A \operatorname{diag}(C^T u)\, B^T, \qquad M_v = A \operatorname{diag}(C^T v)\, B^T \tag{53}$$

For random u, v, the diagonal entries are nonzero, and the ratios $(c_d^T u)/(c_d^T v)$ are distinct, because no two columns of C are parallel. Then

$$M_u M_v^\dagger = A \operatorname{diag}(C^T u) \operatorname{diag}(C^T v)^{-1} A^\dagger \tag{54}$$

so the columns of A are the eigenvectors of $M_u M_v^\dagger$ for its R distinct nonzero eigenvalues, determined up to scaling. Likewise, the columns of B are the eigenvectors of $(M_v^\dagger M_u)^T$, and then C is found by solving a linear system. Both A and B were recovered uniquely (up to scale and the common ordering by eigenvalue matching), so the decomposition is unique. Finally, $\operatorname{rank}(X_{(1)}) = R$ since $X_{(1)} = A(C \odot B)^T$ where A has full column rank and $C \odot B$ has full column rank (a Khatri–Rao product has full column rank when $k_B + k_C \geq R + 1$), so the flattening bound gives $\operatorname{rank}(\mathcal{X}) \geq R$.

This is the model case of how rank is determined in practice: a decomposition gives an upper bound and a flattening gives the matching lower bound. Its weakness is that it needs $R \leq \min\{I, J\}$, so it cannot handle *overcomplete* tensors with rank above the dimension. De Lathauwer's later method based on simultaneous matrix diagonalization extends this to the generic case $R \leq K$ with $R(R-1) \leq I(I-1)J(J-1)/2$, and the simultaneous generalized Schur decomposition of De Lathauwer, De Moor, and Vandewalle gives a more numerically stable alternative when $R \leq \min\{I, J\}$ (both as described in the review).

3.4.2 Numerical Fitting

When none of these structured results applies, the standard practice is the one described in the review: fit a rank-R CP model for $R = 1, 2, 3, \ldots$ with alternating least squares (ALS) and stop at the first R with an exact fit. ALS fixes all factor matrices but one, which makes the problem a linear least squares problem, e.g.

$$\hat{A} = X_{(1)}(C \odot B)(C^T C * B^T B)^\dagger \tag{55}$$

using the Khatri–Rao identity from Lemma 1.1, and then cycles through the modes. It is not guaranteed to converge to even a stationary point, and it depends heavily on the initialization. Derivative-based alternatives (damped Gauss–Newton, PMF3) converge better but are more expensive. There is also no way to *certify* the answer: failing to find an exact fit at rank R does not show that the rank is above R, and with noisy data fit alone can't pick R (Bro and Kiers proposed the CORCONDIA consistency diagnostic for this). And there is a more fundamental reason it can fail, which is the next topic.

3.5 Border Rank and Degeneracy

The question of whether a decomposition can be *found* by numerical optimization runs into a topological problem: the set of tensors of rank $\leq r$ is not closed (for $r \geq 2$ and $N \geq 3$), unlike for matrices.

Definition 3.1 (Border Rank)

The **border rank** of $\mathcal{X}$, denoted $\underline{\text{rank}}(\mathcal{X})$, is the smallest r such that $\mathcal{X}$ is a limit of tensors of rank at most r,

$$\underline{\text{rank}}(\mathcal{X}) \coloneqq \min\left\{r : \forall \epsilon > 0,\ \exists \mathcal{E} \text{ with } \|\mathcal{E}\| < \epsilon,\ \text{rank}(\mathcal{X}+\mathcal{E}) \leq r\right\} \tag{56}$$

Clearly $\underline{\text{rank}}(\mathcal{X}) \leq \text{rank}(\mathcal{X})$.

Example 3.5 (The W Tensor Has Border Rank 2 but Rank 3)

Let $\mathcal{W} = a_1 \circ b_1 \circ c_2 + a_1 \circ b_2 \circ c_1 + a_2 \circ b_1 \circ c_1$ with a_1, a_2 and b_1, b_2 and c_1, c_2 linearly independent. Expanding gives

$$\frac{1}{\epsilon}\Big[(a_1+\epsilon a_2)\circ(b_1+\epsilon b_2)\circ(c_1+\epsilon c_2) - a_1\circ b_1\circ c_1\Big] = \mathcal{W} + \epsilon\big(a_2\circ b_2\circ c_1 + a_2\circ b_1\circ c_2 + a_1\circ b_2\circ c_2\big) + \epsilon^2\, a_2\circ b_2\circ c_2 \tag{57}$$

The left side has rank at most 2 and converges to $\mathcal{W}$ as $\epsilon \to 0$. Since rank-one tensors form a closed set and $\mathcal{W}$ is not rank-one, $\underline{\text{rank}}(\mathcal{W}) = 2$. But by the substitution example, $\text{rank}(\mathcal{W}) = 3$.

This is the example from the review (due to Bini et al. 1979, also in Knuth and Paatero) rewritten, and it exhibits the typical features of degeneracy: two factors become nearly parallel with magnitudes $\to \infty$ and opposite signs. The consequences are the following.

- *Best low-rank approximations may not exist.* De Silva and Lim (2008) show that the set of tensors without a best rank-r approximation has positive Lebesgue measure for some sizes, so this is not a rare event. Stegeman shows that for $I \times I \times 2$ tensors, any tensor of rank $I+1$ has no best rank-I approximation.
- *Rank and border rank can differ* in the context of important problems. For 2×2 matrix multiplication, a $4 \times 4 \times 4$ tensor, both equal 7 (this is Strassen's algorithm, and the border rank was proved to be 7 by Landsberg, 2006). For 3×3, the review lists rank in $[19, 23]$ and border rank in $[14, 21]$.
- *Border rank is also easier to study with geometry*, since it is about the Zariski closure of the set of rank $\leq r$ tensors, namely the rth secant variety of the Segre variety. This is where the Koszul/Young flattenings above live.

4 Typical Ranks

Given that the rank of an individual tensor is hard to compute, we might ask a different question: what rank does a *typical* tensor have?

4.1 Background: The Zariski Topology

To say what "typical" and "generic" mean, we need a few facts from algebraic geometry. Everything here is over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, and the ambient space is $\mathbb{F}^n$, which for us is a space of tensors $\mathbb{F}^{I_1 \times \cdots \times I_N}$ with $n = I_1 \cdots I_N$.

Definition 4.1 (Algebraic Set)

For a set S of polynomials in n variables with coefficients in $\mathbb{F}$, its **zero set** is $V(S) \coloneqq \{x \in \mathbb{F}^n : f(x) = 0 \text{ for all } f \in S\}$. A set of this form is called an **algebraic set** (or an **affine variety**).

Example 4.1

The set of 2×2 matrices of rank at most 1 is $V(\{ad - bc\})$. More generally, the set of $I \times J$ matrices of rank at most r is an algebraic set, since it is the zero set of all $(r+1) \times (r+1)$ minors. By contrast, the unit disk $\{x^2 + y^2 \leq 1\}$ is not algebraic, since algebraic sets are defined by equalities.

Algebraic sets behave like the closed sets of a topology. Indeed, $V(\{1\}) = \emptyset$, $V(\{0\}) = \mathbb{F}^n$, intersections of algebraic sets are algebraic since $\bigcap_i V(S_i) = V(\bigcup_i S_i)$, and finite unions are algebraic since $V(f) \cup V(g) = V(fg)$.

Definition 4.2 (Zariski Topology)

The **Zariski topology** on $\mathbb{F}^n$ is the topology whose closed sets are the algebraic sets. Its open sets are their complements, the **Zariski-open** sets. The **Zariski closure** $\overline{A}$ of a set A is the smallest algebraic set containing it, and A is **Zariski-dense** if $\overline{A} = \mathbb{F}^n$.

This topology is extremely coarse. For example, on $\mathbb{C}$ the proper algebraic sets are exactly the finite sets, so the nonempty Zariski-open sets are the sets with finite complement. In particular, it is not Hausdorff. The point is that a proper algebraic set is *small*, and so a nonempty Zariski-open set is *large*. This is made precise by the following facts.

Lemma 4.1 (Zariski-Open Sets Are Large)

Let f be a nonzero polynomial on $\mathbb{F}^n$.

1. The zero set $V(f)$ has Lebesgue measure zero. Consequently every nonempty Zariski-open set has a complement of measure zero, and so it is dense in the usual (Euclidean) topology.
2. (*Irreducibility of* $\mathbb{F}^n$) Any two nonempty Zariski-open sets intersect (and so their intersection is again nonempty and open).
3. The real points $\mathbb{R}^n$ are Zariski-dense in $\mathbb{C}^n$, i.e. a polynomial that vanishes on all of $\mathbb{R}^n$ is the zero polynomial.

Proof. (1) We induct on n. For $n = 1$, a nonzero polynomial has finitely many roots. For $n > 1$, write $f = \sum_k c_k(x_1, \ldots, x_{n-1})\, x_n^k$ where some c_k is a nonzero polynomial. By induction, $V(c_k) \subset \mathbb{F}^{n-1}$ has measure zero, and for every x' outside it, $f(x', \cdot)$ is a nonzero polynomial in one variable, with finitely many roots. By Fubini, $V(f)$ has measure zero. (For $\mathbb{F} = \mathbb{C}$, apply this to the real or imaginary part of f as a polynomial in the $2n$ real variables, one of which is nonzero.) If U is a nonempty Zariski-open set, then its complement is $V(S)$ for some S containing a nonzero polynomial f, and so $V(S) \subseteq V(f)$ has measure zero. A set whose complement has measure zero is dense in the usual topology.
(2) A nonempty Zariski-open set contains a set of the form $\{f \neq 0\}$ with f a nonzero polynomial. Two such sets $\{f \neq 0\}$ and $\{g \neq 0\}$ intersect in $\{fg \neq 0\}$, which is nonempty since fg is a nonzero polynomial and so by (1) is not identically zero.
(3) If f vanishes on $\mathbb{R}^n$ then f vanishes on a set of positive measure in $\mathbb{R}^n$. Writing $f = f_1 + if_2$ with f_1, f_2 real polynomials, each vanishes on $\mathbb{R}^n$, and by (1) applied over $\mathbb{R}$ each must be zero.

Part (2) is what makes the word "generic" meaningful. A generic property is one that holds on a nonempty

Zariski-open set, and part (2) says that any two generic properties hold simultaneously on a generic set. In particular, "the complement of a proper algebraic set" is the right notion of "almost everywhere" for polynomial problems.

Next, we need to talk about sets that are the image of a polynomial map, since the set of tensors of rank at most r is the image of the map $(a_d^{(n)}) \mapsto \sum_{d=1}^r a_d^{(1)} \circ \cdots \circ a_d^{(N)}$. Such sets need not be closed in either topology.

Definition 4.3 (Irreducible, Dimension)

An algebraic set X is **irreducible** if it is not the union of two proper algebraic subsets. Lemma 4.1 (2) says that $\mathbb{F}^n$ is irreducible. The **dimension** of an irreducible algebraic set is the dimension of the manifold of its smooth points (equivalently, the dimension of its tangent space at a generic point).

The following standard facts about images of polynomial maps are the ones we use. They hold over $\mathbb{C}$.

Theorem 4.2 (Images of Polynomial Maps)

Let $\varphi : \mathbb{C}^M \to \mathbb{C}^N$ be a polynomial map.

1. The Zariski closure $\overline{\varphi(\mathbb{C}^M)}$ is irreducible.
2. (*Chevalley*) $\varphi(\mathbb{C}^M)$ contains a nonempty Zariski-open subset of its closure. In particular, if $\varphi(\mathbb{C}^M)$ is Zariski-dense in $\mathbb{C}^N$, then it contains a nonempty Zariski-open set.
3. $\dim \overline{\varphi(\mathbb{C}^M)} = \max_x \operatorname{rank} D\varphi(x)$, the rank of the Jacobian at a generic point; in particular, $\dim \leq M$. Therefore, $\varphi(\mathbb{C}^M)$ is Zariski-dense if and only if $D\varphi(x)$ has rank N at a generic x.
4. The Zariski closure of $\varphi(\mathbb{C}^M)$ is also its closure in the usual topology.

Part (1) follows since the closure of the image of an irreducible set is irreducible, and $\mathbb{C}^M$ is irreducible. Part (3) is Terracini's lemma in the form we use, and it is what makes the Jacobian method for computing generic ranks work. Part (4) says that over $\mathbb{C}$, the closure of a set of tensors, e.g. those of rank at most r, is the same in both topologies, so the border rank is also the smallest r for which a tensor lies in the Zariski closure of the tensors of rank at most r.

Finally, over $\mathbb{R}$ the situation is different, since the image of a real polynomial map is not an algebraic set or even its open subset. It is only a **semialgebraic set**, i.e. a finite union of sets defined by polynomial equalities and inequalities (Tarski–Seidenberg). For these, positive Lebesgue measure is equivalent to having a nonempty interior, which is why a typical rank over $\mathbb{R}$ is a rank attained on an open set. Note the contrast with $\mathbb{C}$. A real polynomial equation cuts out a set of measure zero, but a real polynomial *inequality* like $\Delta(\mathcal{X}) > 0$ in the $2 \times 2 \times 2$ case cuts out an open set, and each of these can be a typical rank.

4.2 Generic and Typical Properties

To make this precise, we have to say what "typical" means, and it turns out that we don't need to pick a specific probability distribution.

Definition 4.4 (Generic and Typical Properties)

A property of tensors in $\mathbb{C}^{I \times J \times K}$ is **generic** if it holds outside the zero set of a nonzero polynomial in the entries (i.e. on a Zariski-open dense set). A property of tensors in $\mathbb{R}^{I \times J \times K}$ is **typical** if it holds on a set of positive Lebesgue measure.

The zero set of a nonzero polynomial has Lebesgue measure zero, so a generic property holds with probability one for *any* distribution on the entries that has a density with respect to Lebesgue measure, e.g. iid Gaussian entries. All such distributions have the same null sets, so which ranks occur with probability zero, and which

occur with positive probability, doesn't depend on the choice. What does depend on it are the actual probabilities. For example, the 79% and 21% for rank 2 and rank 3 of $2 \times 2 \times 2$ tensors came from Gaussian entries, and a different density would give different percentages, but both ranks would still occur with positive probability.

Over $\mathbb{C}$, the set of tensors of rank at most r has an irreducible closure (it is the closure of the image of a polynomial map), so there is a single rank r for which the closure first becomes everything. This is the **generic rank**. Over $\mathbb{R}$ there can be several typical ranks, as we saw for $2 \times 2 \times 2$, and this is what makes the real case interesting. In this section we collect the main general theorems about typical ranks. The results for tensors of specific shapes are in the next section.

4.3 The Smallest Typical Rank

Over $\mathbb{R}$ the first question is how the typical ranks over $\mathbb{R}$ relate to the generic rank over $\mathbb{C}$. Since real tensors are also complex tensors, and a real decomposition is also a complex one, $\operatorname{rank}_{\mathbb{R}}(\mathcal{X}) \geq \operatorname{rank}_{\mathbb{C}}(\mathcal{X})$. The following says that the smallest typical rank is exactly the complex generic rank.

Theorem 4.3 (Smallest Typical Rank)

Let g be the generic rank of $\mathbb{C}^{I_1 \times \cdots \times I_N}$. Then g is a typical rank over $\mathbb{R}$, and every typical rank over $\mathbb{R}$ is at least g.

Proof. Let $\varphi_r : (a_d^{(n)}) \mapsto \sum_{d=1}^{r} a_d^{(1)} \circ \cdots \circ a_d^{(N)}$ be the parametrization of tensors of rank at most r, which is a polynomial map with real coefficients.
No typical rank below g. By definition of g, for $r < g$ the closure of the image of φ_r over $\mathbb{C}$ is a proper subvariety of $\mathbb{C}^{I_1 \times \cdots \times I_N}$, so it is the zero set of a nonzero polynomial and has measure zero. Since the real rank is at least the complex rank, almost every real tensor has real rank at least g.
g is typical. By definition of the generic rank, the image of φ_g over $\mathbb{C}$ is dense, which (by generic smoothness, in characteristic zero) means that the differential of φ_g is surjective at a generic point. That is, some maximal minor of the Jacobian of φ_g is a nonzero polynomial in the parameters. This polynomial has real coefficients, and a nonzero polynomial cannot vanish on all of $\mathbb{R}^M$, so there is a *real* parameter point where the differential of $\varphi_g : \mathbb{R}^M \to \mathbb{R}^{I_1 \cdots I_N}$ is surjective. By the implicit function theorem, the image of φ_g then contains an open ball, which has positive Lebesgue measure. So a set of positive measure of real tensors has real rank at most g, hence exactly g by the first part.

This is the reason the Jacobian method of Comon et al. (next section) computes the smallest typical rank over $\mathbb{R}$. Notice that the proof says nothing about *other* typical ranks, which exist when the image of φ_g over $\mathbb{R}$ is not everything.

4.4 Expected Typical Rank and Defective Cases

To find g, we count dimensions.

Lemma 4.4 (Expected Generic Rank)

The set of tensors of rank at most r in $\mathbb{C}^{I \times J \times K}$ has dimension at most $r(I+J+K-2)$. So the generic rank is at least

$$\left\lceil \frac{IJK}{I+J+K-2} \right\rceil \tag{58}$$

with equality for most, but not all, sizes.

Proof. We first count the dimension of the set of rank-one tensors. A rank-one tensor $a \circ b \circ c$ is determined by $(a, b, c) \in \mathbb{C}^I \times \mathbb{C}^J \times \mathbb{C}^K$, which has $I + J + K$ parameters. But different triples can give the same tensor, since we can move scalars between the factors,

$$(\alpha a) \circ (\beta b) \circ (\gamma c) = (\alpha\beta\gamma)\, a \circ b \circ c, \qquad \alpha, \beta, \gamma \neq 0 \tag{59}$$

so $(\alpha a, \beta b, \gamma c)$ gives the same tensor as (a, b, c) whenever $\alpha\beta\gamma = 1$. This is a 2-parameter family, since we can pick α, β freely and then $\gamma = 1/(\alpha\beta)$. Conversely, if $a \circ b \circ c = a' \circ b' \circ c' \neq 0$, then comparing fibers shows that $a' = \alpha a$, $b' = \beta b$, $c' = \gamma c$ for some scalars with $\alpha\beta\gamma = 1$. So the fiber of the map $(a, b, c) \mapsto a \circ b \circ c$ over each nonzero rank-one tensor is exactly this 2-dimensional family, and the set of rank-one tensors has dimension

$$(I + J + K) - 2 \tag{60}$$

As a sanity check, projectively the rank-one tensors form the product $\mathbb{P}^{I-1} \times \mathbb{P}^{J-1} \times \mathbb{P}^{K-1}$ (the Segre variety) of dimension $(I-1) + (J-1) + (K-1) = I + J + K - 3$, and the set of rank-one tensors is its cone, which has one more dimension.
Now a tensor of rank at most r is a sum of r rank-one tensors, so the set of such tensors is the image of the sum map from r copies of the set of rank-one tensors, which has dimension $r(I + J + K - 2)$. The image of a polynomial map cannot have larger dimension than its domain, so the set of tensors of rank at most r has dimension at most $r(I + J + K - 2)$. For this set to fill up all of $\mathbb{C}^{I \times J \times K}$, which has dimension IJK, we need $r(I + J + K - 2) \geq IJK$, which is the claimed bound.

The format is called **defective** if g is strictly larger than the expected value, and for most formats this does not happen.

Theorem 4.5 (Lickteig, 1985)

The generic rank (hence the smallest typical rank) of $n \times n \times n$ tensors is $\lceil n^3/(3n-2) \rceil$ for every $n \neq 3$, whereas for $n = 3$ it is 5 and not 4.

Proof. The lower bound is the dimension count above. For the upper bound, one has to show that for $r = \lceil n^3/(3n-2) \rceil$ there is a choice of parameters where the Jacobian of φ_r has full rank n^3, i.e. that the tangent spaces $T_1, \ldots, T_r$ to the set of rank-one tensors at r suitable points span the whole space (this is Terracini's lemma), and Lickteig shows that this can be done for every $n \neq 3$. The case $n = 3$ is the exception because the dimension count is not attained: although $r = 4$ gives $4 \cdot 7 = 28 \geq 27$ parameters, the closure of the set of tensors of rank at most 4 is a hypersurface (Strassen's equation) and not all of $\mathbb{C}^{3\times3\times3}$. So the generic rank is 5.

Strassen showed that the formats $3 \times n \times n$ with n odd are also defective, and $3 \times 3 \times 3$ is the smallest case. This is why the table of typical ranks has so many entries that don't match the naive count.

4.5 The Set of Typical Ranks

Over $\mathbb{R}$ there can be several typical ranks, so we may ask what the set of typical ranks looks like.

Theorem 4.6 (Bernardi, Blekherman, Ottaviani, 2018)

The typical ranks over $\mathbb{R}$ of a given format form a set of consecutive integers. That is, if g is the smallest typical rank and m is the largest, then every integer in $[g, m]$ is a typical rank.

This is consistent with all of the examples we have seen: $\{2, 3\}$ for $2 \times 2 \times 2$, $\{3, 4\}$ for $3 \times 3 \times 2$, $\{4, 5\}$ for $4 \times 4 \times 2$, and a single value for tall tensors. The result is proven for ranks with respect to any real

projective variety, not only the Segre variety, and so it also applies to symmetric tensors. For instance, the typical symmetric ranks of ternary cubics are only $\{4\}$, of quaternary cubics $\{5, 6\}$, and of ternary quartics include 6 and 7. We omit the proof.

The next question is how far apart the smallest and largest typical ranks can be.

Theorem 4.7 (Blekherman and Teitler, 2015)

The maximum real rank, and in particular the largest typical rank, of tensors of a given format is at most $2g$, where g is the smallest typical rank (the generic rank over $\mathbb{C}$).

Proof. The idea is that any tensor is the sum of two tensors of rank at most g. Over $\mathbb{C}$, this is easy: the set V of tensors of rank at most g contains a Zariski-open dense set, so for any tensor $\mathcal{T}$, both V and $\mathcal{T} - V$ contain dense open sets. These must intersect, giving $\mathcal{S}$ with $\mathcal{S} \in V$ and $\mathcal{T} - \mathcal{S} \in V$, so $\mathcal{T} = \mathcal{S} + (\mathcal{T} - \mathcal{S})$ has rank at most $2g$. Over $\mathbb{R}$, the set of real tensors of real rank at most g is not dense, so this argument needs more care, and that is the content of the paper.

So for the $2 \times 2 \times 2$ case $g = 2$ and the largest typical rank $3 \leq 4$.

4.6 Typical Ranks of Specific Shapes

Beyond these general facts, the typical ranks of specific formats have been worked out. The results we prove in the next section are the following.

- $2 \times 2 \times 2$: typical ranks $\{2, 3\}$ (Theorem 6.1).
- $I \times J \times 2$: $\{I, I + 1\}$ if $I = J$, I if $J < I < 2J$, and $2J$ if $I \geq 2J$ (Theorem 6.3).
- Tall tensors $I \times J \times K$ with $JK - J < I < JK$: typical rank I (Theorem 6.5).
- Tensors with symmetric slices (Lemma 6.7).

A more recent result extends the tall case to a wider range of formats, and it also has a nice geometric proof.

Theorem 4.8 (Breiding, Eggleston, Rosana, 2024)

Let $\mathcal{X} \in \mathbb{R}^{m \times n \times \ell}$ with $(m-1)(n-1) + 1 \leq \ell \leq mn$. Then ℓ is always a typical rank, and all typical ranks lie in $\{\ell, \ell + 1\}$.

Proof. By the Slice-Span Characterization, if the ℓ frontal slices are linearly independent and span $L \subseteq \mathbb{R}^{m \times n}$, then $\operatorname{rank}(\mathcal{X})$ is the smallest number of rank-one matrices whose span contains L. The flattening bound gives $\operatorname{rank}(\mathcal{X}) \geq \ell$, and equality holds exactly when L is spanned by rank-one matrices *lying in* L. So we look at the intersection of L with the cone of rank-one matrices (the cone over the Segre variety, of dimension $m + n - 1$). Since $\dim L = \ell$ in the ambient space of dimension mn, the expected dimension of the intersection is

$$\ell + (m + n - 1) - mn \geq 1 \iff \ell \geq (m-1)(n-1) + 1 \tag{61}$$

which is exactly the range in the theorem, so L meets the Segre variety in a positive-dimensional set. Whether L is spanned by its *real* rank-one points is an open condition, which holds for a set of positive measure, and this gives rank ℓ as typical. The rank probabilities can then be expressed in terms of the distribution of the intersection points of a random linear space with the Segre variety. We omit the proof that no other ranks than ℓ and $\ell + 1$ occur.

For example, the typical ranks of real $3 \times 3 \times 5$ tensors are $\{5, 6\}$, in agreement with the numerical tables.

5 Symmetric Rank

Definition 5.1 (Symmetric Rank)

The **symmetric rank** of a supersymmetric tensor $\mathcal{X} \in \mathbb{C}^{I \times \cdots \times I}$ is

$$\operatorname{rank}_S(\mathcal{X}) \coloneqq \min \left\{ R : \mathcal{X} = \sum_{r=1}^{R} a_r \circ a_r \circ \cdots \circ a_r \right\} \tag{62}$$

Via the identification of supersymmetric tensors with homogeneous polynomials, this is the *Waring rank*: the smallest number of dth powers of linear forms that sum to a given form of degree d.

Theorem 5.1 (Alexander–Hirschowitz, 1995)

A generic form of degree $d \geq 3$ in n variables over $\mathbb{C}$ has symmetric rank

$$\left\lceil \frac{1}{n} \binom{n+d-1}{d} \right\rceil \tag{63}$$

except for $(d, n) \in \{(3,5), (4,3), (4,4), (4,5)\}$, where the rank is one more.

Outline of proof. The rth secant variety of the Veronese variety has expected dimension $\min\{rn, \binom{n+d-1}{d}\}$ (projectively, minus one), and the theorem says this is the actual dimension except in the listed cases. Equivalently, one needs that general r points in $\mathbb{P}^{n-1}$ impose independent conditions on degree-d hypersurfaces that are *double* at these points (Terracini's lemma turns the tangent space of the secant variety at a sum of r powers into the space of forms vanishing doubly at the r points). Alexander and Hirschowitz prove this by a long induction on n and d by specializing points onto a hyperplane.

For a *specific* form, the classical result is for $n = 2$ and is a rank-determination algorithm in the same spirit as Theorem 3.8.

Theorem 5.2 (Sylvester, 1851)

A binary form f of degree d can be written $f = \sum_{i=1}^{r} \ell_i^d$ with distinct linear forms ℓ_i if and only if there is a form g of degree r with r distinct roots such that $g(\partial) f = 0$.

Outline of proof. Using apolarity, if $g = \prod_i (\beta_i x - \alpha_i y)$ and $\ell_i = \alpha_i x + \beta_i y$, then $g(\partial)\ell_i^d = 0$ for each i (differentiating a power of ℓ_i by the form vanishing at ℓ_i kills it), so $g(\partial)$ annihilates any combination of the ℓ_i^d. Conversely, the annihilator condition $g(\partial) f = 0$ is a linear system for the coefficients of g whose matrix is the *catalecticant* (Hankel) matrix of f, and the roots of a kernel vector g give the ℓ_i. So $\operatorname{rank}_S(f)$ is found by looking for the smallest r with a kernel vector of the catalecticant with distinct roots. Comon and Mourrain and later Brachat, Comon, Mourrain, and Tsigaridas generalized this to several variables.

5.1 Comon's Conjecture

A natural question is whether imposing symmetry costs anything: is $\operatorname{rank}_S(\mathcal{X}) = \operatorname{rank}(\mathcal{X})$ for a supersymmetric tensor? This is **Comon's conjecture**, and it is known in several special cases (e.g. matrices, and small ranks). Shitov (2018) announced a counterexample (an $800 \times 800 \times 800$ symmetric tensor of rank at most 903 whose symmetric rank is larger), but an erratum has since been published saying that Shitov found a serious error in the argument, and it is not currently known whether it can be corrected. A recent preprint by Lovitz claims a $27 \times 27 \times 27$ counterexample over $\mathbb{Q}$ (rank 55, symmetric rank 56), though this has not

yet been peer reviewed.[2]

[2] https://arxiv.org/abs/1705.08740, https://epubs.siam.org/doi/full/10.1137/23M1623781, https://arxiv.org/abs/2609.28292v1.

6 Tensors of Specific Shapes

Even though there is no general algorithm for computing the rank, the rank is completely understood for tensors of certain shapes. The tables in the review (maximum ranks of $I \times J \times 2$ and $3 \times 3 \times 3$ tensors, typical ranks of $2 \times 2 \times 2$, $3 \times 3 \times 2$, $I \times J \times 2$, tall tensors, tensors with symmetric slices, etc.) are summaries of a handful of papers, mostly from the psychometrics literature. In this section we go through them, with the main statement and an outline of the proof. We will use the slice-span characterization and the rank of a regular pencil throughout.

6.1 Tensors with Two Slices: $2 \times 2 \times 2$

The first result of this kind is due to Kruskal, who noticed that the set of $2 \times 2 \times 2$ tensors of rank less than the maximum rank has positive volume (this is what makes the typical rank not unique). The proof that was written down by Ten Berge in 1991 is the following.[3] For a tensor $\mathcal{X} \in \mathbb{R}^{2\times 2\times 2}$ with frontal slices X_1, X_2, define

$$\Delta(\mathcal{X}) \coloneqq \text{discriminant of the quadratic } \lambda \mapsto \det(X_2 - \lambda X_1) \tag{64}$$

which is a homogeneous polynomial of degree 4 in the entries, known as Cayley's **hyperdeterminant**, and it is also called Kruskal's polynomial.

Theorem 6.1 (Kruskal; Ten Berge)

Let $\mathcal{X} \in \mathbb{R}^{2\times 2\times 2}$ be nonzero.

1. $\operatorname{rank}(\mathcal{X}) \le 3$.
2. $\operatorname{rank}(\mathcal{X}) = 1$ if and only if all six of its slices (in all three directions) are singular, unless $\mathcal{X}$ is **superdiagonal**, i.e. has exactly two nonzero entries that never lie in the same slice, which has rank 2.
3. Suppose that some slice is nonsingular, and by permuting the directions and slices say that it is X_1. If X_2 is a scalar multiple of X_1, then the rank is 2. Otherwise,

$$\operatorname{rank}(\mathcal{X}) = \begin{cases} 2 & \text{if } \Delta(\mathcal{X}) > 0 \\ 3 & \text{if } \Delta(\mathcal{X}) \le 0 \end{cases} \tag{65}$$

   Over $\mathbb{C}$ the condition becomes $\Delta(\mathcal{X}) \neq 0$ for rank 2.

In particular both 2 and 3 are typical ranks over $\mathbb{R}$, and the generic rank over $\mathbb{C}$ is 2.

Proof. By Lemma 3.1, $\operatorname{rank}(\mathcal{X})$ is the smallest n such that $X_1 = ADB^T$ and $X_2 = AEB^T$ with $A, B \in \mathbb{R}^{2\times n}$ and D, E diagonal $n \times n$.
Part 2. If the rank is one, then $X_1 = d\,ab^T$ and $X_2 = e\,ab^T$, so both slices (and by symmetry all six) are singular. Conversely, if all slices are singular and some slice is zero, then the other slice is ab^T and $\mathcal{X} = a \circ b \circ e_1$. If no slice is zero, write $X_1 = ab^T$ with $b = (1, \lambda)^T$, so that $X_1 = [a \,|\, \lambda a]$ in terms of columns, and then singularity of the second column slice and of the horizontal slices force the columns of X_2 to be μa and $\lambda\mu a$, giving $\mathcal{X} = a \circ (1, \lambda)^T \circ (1, \mu)^T$ with rank one. The only exception is when the argument ends with $X_1 = [a \,|\, 0]$ and $X_2 = [0 \,|\, d]$, which is superdiagonal, and it has rank 2 since X_1, X_2 are not scalar multiples of the same rank-one matrix.
Part 1. If every slice is singular then the rank is at most 2 by Part 2. Otherwise, a nonsingular slice can be taken to be X_1. Let $Y = X_2 X_1^{-1} = (y_{ij})$. By multiplying on the right by X_1^{-1} (a change of

[3] J. M. F. Ten Berge, *Kruskal's polynomial for* $2 \times 2 \times 2$ *arrays and a generalization to* $2 \times n \times n$ *arrays*, Psychometrika 56 (1991). The exposition of Bremner and Hu, *On Kruskal's theorem that every* $3 \times 3 \times 3$ *array has rank at most 5*, also contains complete proofs of these facts.

basis), it suffices to decompose the pair (I, Y). Define

$$A = \begin{bmatrix} 1 & 0 & y_{12} \\ 0 & 1 & y_{21} \end{bmatrix}, \quad D = \operatorname{diag}(1, 1, 0), \quad E = \operatorname{diag}(y_{11} - y_{12},\ y_{22} - y_{21},\ 1), \quad B^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} X_1 \tag{66}$$

Then a direct computation gives $ADB^T = X_1$ and $AEB^T = YX_1 = X_2$, so there is a decomposition with three terms.

Part 3. Suppose X_1 is nonsingular and X_2 is not a scalar multiple of X_1. By Part 1 the rank is 2 or 3, and by Theorem 3.8, it is 2 if and only if $X_1^{-1}X_2$ is diagonalizable over the field, i.e. (since X_2 is not a multiple of X_1) if and only if $\det(X_2 - \lambda X_1)$ has two distinct roots in the field. Concretely, if $X_1 = d_1 a_1 b_1^T + d_2 a_2 b_2^T$ and $X_2 = e_1 a_1 b_1^T + e_2 a_2 b_2^T$, then $X_2 - (e_j/d_j)X_1$ is singular for $j = 1, 2$, and these two values of λ are distinct. Conversely, if $\lambda_1 \neq \lambda_2$ are roots, then the matrices $X_2 - \lambda_j X_1$ are singular and nonzero, hence rank one, and we can write

$$u_1 v_1^T = \frac{X_2 - \lambda_2 X_1}{\lambda_1 - \lambda_2}, \qquad u_2 v_2^T = -\frac{X_2 - \lambda_1 X_1}{\lambda_1 - \lambda_2} \tag{67}$$

which satisfy $X_1 = u_1 v_1^T + u_2 v_2^T$ and $X_2 = \lambda_1 u_1 v_1^T + \lambda_2 u_2 v_2^T$. Two distinct real roots exist if and only if the discriminant $\Delta(\mathcal{X})$ is positive, and two distinct complex roots if and only if $\Delta(\mathcal{X}) \neq 0$.

Finally, $\{\Delta > 0\}$ and $\{\Delta < 0\}$ are both nonempty open sets (e.g. $X_1 = I$ and $X_2 = \operatorname{diag}(1, 2)$ versus $X_2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$), so ranks 2 and 3 both occur with positive probability.

When $\Delta(\mathcal{X}) = 0$ and X_2 is not proportional to X_1, the rank is 3 even though such a tensor is a limit of rank-two tensors. For example $X_1 = I$ and $X_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has rank 3 (since the Jordan block is not diagonalizable), but it is the limit as $a \to 0$ of $(I, \begin{bmatrix} 0 & 1 \\ a^2 & 0 \end{bmatrix})$ which has $\Delta = 4a^2 \neq 0$ and hence rank 2. This is again the example of border rank two but rank three. Monte Carlo experiments with Gaussian entries give about 79% rank two and 21% rank three.

6.2 Tensors with Two Slices: $I \times I \times 2$ and $I \times J \times 2$

The same idea generalizes to all pencils, and this is what is behind the $K = 2$ entries in the review's tables. The first result is about the maximum rank.

Theorem 6.2 (Kruskal; JáJá)

The maximum rank of an $I \times J \times 2$ tensor is $\min\{I, J\} + \min\{I, J, \lfloor \max\{I, J\}/2 \rfloor\}$. In particular, the maximum rank of an $I \times I \times 2$ tensor is $\lfloor 3I/2 \rfloor$.

Proof. We prove the upper bound for $I \times I \times 2$ tensors whose pencil is regular, i.e. such that after a change of basis in mode 3 the first slice X_1 is invertible. The general case is handled in the same way with the Kronecker canonical form of the pencil. By a change of basis in modes $1, 2$, we may replace (X_1, X_2) by (I, C) where $C = X_1^{-1}X_2$ is in rational (or Jordan) canonical form, which is block diagonal with blocks $C_1, \ldots, C_m$ of sizes $s_1, \ldots, s_m$, each the companion matrix of a polynomial. Then (I, C) is the direct sum of the pairs (I_{s_i}, C_i), so the rank is at most the sum of the ranks.

Each block satisfies $\operatorname{rank}(I_s, C_s) \leq s + 1$, and $= s$ if $s = 1$. To see this, write the companion matrix as $C_s = C' + D$, where D is the rank-one matrix that agrees with C_s on the last column and is zero elsewhere, and C' is the companion matrix of a polynomial with s distinct real roots (any monic polynomial is a companion matrix after changing the last column, so we are free to choose these roots).

Then

$$(I_s, C_s) = (0, D) + (I_s, C') \tag{68}$$

where $(0, D)$ is rank one, and (I_s, C') has rank s by Theorem 3.8 since C' is diagonalizable.
So the rank is at most $n + \#\{i : s_i \geq 2\} \leq n + \lfloor n/2 \rfloor$, since blocks of size 1 are free and there are at most $\lfloor n/2 \rfloor$ blocks of size at least 2.

This explains $2 \times 2 \times 2 \mapsto 3$, $3 \times 3 \times 2 \mapsto 4$ and $4 \times 4 \times 2 \mapsto 6$. Next we turn to the typical rank.

Theorem 6.3 (Ten Berge and Kiers)

The typical ranks of $I \times J \times 2$ tensors over $\mathbb{R}$ are the following.

1. If $I = J$, then the typical ranks are $\{I, I + 1\}$.
2. If $J < I < 2J$, then the typical rank is I.
3. If $I \geq 2J$, then the typical rank is $2J$.

Proof. Part 1. With probability one, X_1 is invertible, so $\text{rank}(\mathcal{X}) \geq \text{rank}(X_1) = I$. Let $Y = X_1^{-1} X_2$, which has distinct eigenvalues with probability one, and hence is similar to the companion matrix of its characteristic polynomial.

- If all eigenvalues of Y are real, which is an open condition and so has positive probability, then Y is diagonalizable and the rank is I by Theorem 3.8.
- If Y has a pair of complex eigenvalues, which is also an open condition, then Y is not diagonalizable over $\mathbb{R}$, so the rank is more than I by Theorem 3.8.
- In both cases the rank is at most $I+1$ by the companion matrix argument in the proof of Theorem 6.2, applied to a single block.

Hence rank I and rank $I + 1$ both occur with positive probability, and no other value does.

Proof. Parts 2 and 3. These are special cases of the tall tensor theorem proven below with $K = 2$.

Ten Berge's empirical check of how large the set of rank-n tensors is: for pairs of $n \times n$ Gaussian matrices, the fractions of 100 samples for which $X_2 X_1^{-1}$ had n distinct real eigenvalues were $76, 55, 22, 16, 2, 1, 0$ for $n = 2, \ldots, 8$. So rank n is typical but it becomes increasingly rare. Since the typical ranks in these cases are not unique, we can see why the review lists the typical rank as $\{3, 4\}$ for $3 \times 3 \times 2$, and $\{I, I + 1\}$ for $I \times I \times 2$.

The same technique shows that the typical and maximum rank can be far apart.

Example 6.1 ($4 \times 4 \times 2$)

The maximum rank of $4 \times 4 \times 2$ tensors is 6, but the typical rank is $\{4, 5\}$. For instance, take $X_1 = I_4$ and

$$X_2 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -b \\ 0 & 0 & b & 0 \end{bmatrix}, \qquad b \neq 0 \tag{69}$$

whose eigenvalues $\pm i, \pm bi$ are complex, so the rank is at least 5. For $b^2 \neq 1$, subtracting a suitable rank-one matrix xy^T from X_2 gives a matrix with the real distinct eigenvalues $\pm 1, \pm b$, so the rank is exactly 5. When $b = \pm 1$ this construction is not possible, and the rank is 6 (shown by Thijsse). So the rank is 5 with probability one when b is random, but the maximal rank 6 occurs on a set of measure zero.

6.3 $3 \times 3 \times 3$ Tensors

The case that is not a pencil but is still completely understood is $3 \times 3 \times 3$.

Theorem 6.4 (Kruskal)

Every $3 \times 3 \times 3$ tensor over $\mathbb{R}$ or $\mathbb{C}$ has rank at most 5. Combined with the fact that the generic rank of $3 \times 3 \times 3$ tensors over $\mathbb{C}$ is 5, the maximum rank and typical rank are both 5.

Proof. This was stated by Kruskal in 1989 without proof, and a complete proof was written down by Bremner and Hu (2010), following a simplification by Rocci.[a] Let $\mathcal{T} = [A \mid B \mid C]$ be the tensor with frontal slices A, B, C. The lower bound for the generic rank is Table 2 of Comon et al., discussed later; we outline the upper bound.

Reduction to two singular slices. If some slice is zero, then $\mathcal{T}$ is a $3 \times 3 \times 2$ tensor and has rank at most 4 by Theorem 6.2. Otherwise, if at least two slices are nonsingular, say C is nonsingular, then $\det(A - \lambda C)$ is a cubic polynomial in λ with leading coefficient $\pm \det C \neq 0$. Since the degree is odd, it has a real root, so replacing A by $A - \lambda C$ (a change of basis in mode 3) makes it singular. We do the same for B. So we may assume that A and B are singular.

Reduction to slices of rank at least two. If some slice has rank ≤ 1, say A, then $\mathcal{T} = [A|0|0] + [0|B|C]$ where the first term is rank one and the second is a $3 \times 3 \times 2$ tensor of rank at most 4. So we may assume A, B have rank exactly 2, and C has rank 2 or 3.

Case analysis. We use two lemmas of Kruskal, whose proofs are elementary.

- **One-Edge Lemma.** If two slices D, E have a common null vector, $Dx = Ex = 0$ (or $D^T x = E^T x = 0$), then $\operatorname{rank}(\mathcal{T}) \leq 5$. *Proof:* change basis in mode 2 so that $x = e_1$. Then the first column of D and E vanishes, so $\mathcal{T}$ is the sum of a tensor that is nonzero only in the first column of the third slice (rank one), and a tensor of size $3 \times 2 \times 3$, which has rank at most 4 as a (transposed) $3 \times 3 \times 2$ tensor.
- **Two-Edge Lemma.** If two slices D, E have $Dx = 0$, $y^T D = 0$ and $y^T E x \neq 0$, then $\operatorname{rank}(\mathcal{T}) \leq 5$. *Proof:* choose bases with x, y as first vectors so that D has zero first row and column, and the $(1,1)$ entry $\alpha = y^T E x$ of E is nonzero. Then the first row and column of E (and, after replacing C by $C + E$ if necessary, of C) can be matched by a rank-one matrix $\alpha^{-1} u v^T$. Subtracting these two rank-one tensors leaves a $2 \times 2 \times 2$ tensor, which has rank at most 3 by Theorem 6.1.

If A, B, C all have rank 2, take the null vectors x_i and left null vectors y_i of the three slices and form $X = [x_1|x_2|x_3]$, $Y = [y_1|y_2|y_3]$, so that $Y^T[A|B|C]X$ has a zero pattern. If the Two-Edge Lemma applies, we are done. If not, then all the diagonal entries $y_i^T M x_i$ vanish, and we split into cases on the ranks of X and Y. When two columns of X are dependent, we use the One-Edge Lemma; when X has rank 2 one finds that the first rows of the three slices are dependent and removes them to get a $2 \times 3 \times 3$ tensor of rank at most 4; and when both X, Y are invertible one subtracts three explicit rank-one tensors to be left with two more. When C is invertible, a similar case analysis applies, now using the cases $\alpha A + \beta B + C$ singular or not. Each case exhibits a decomposition with at most five terms.

[a] `http://www.cecm.sfu.ca/~jha107/BH333rank5v7.pdf`

Over a field that is not algebraically closed or $\mathbb{R}$, the statement can fail: over $\mathbb{F}_2$, there exist $3 \times 3 \times 3$ arrays of rank 6 (von zur Gathen). The odd degree 3 is used in the *reduction to two singular slices* step, where over $\mathbb{R}$ we need the cubic $\det(A - \lambda C)$ to have a real root.

6.4 Tall Tensors

The case where one dimension is large compared to the others was settled by Ten Berge. Let $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$ with $I \geq J \geq K$. We call it

- **very tall** if $I \geq JK$,

- **tall** if $JK - J < I < JK$,
- **compact** if $I \leq JK - J$.

Theorem 6.5 (Ten Berge, 2000)

Let $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$ with $I \geq J \geq K$.
1. If $\mathcal{X}$ is very tall, then the typical rank (and maximum rank) is JK.
2. If $\mathcal{X}$ is tall, then the typical rank is I.
3. If $I = JK - J$, then the rank is I or $I + 1$ almost surely, and rank I occurs with positive probability. Moreover, if $K > 2$ and J is odd then the typical rank is exactly I.

Proof. Part 1. Let $X_1, \dots, X_K \in \mathbb{R}^{I \times J}$ be the frontal slices. Almost surely, the $I \times JK$ matrix $[X_1 | \cdots | X_K]$ has rank JK, so by the flattening bound $\text{rank}(\mathcal{X}) \geq JK$. For the upper bound, write $X_k = A_k B_k^T$ with $A_k = X_k$ and $B_k = I_J$, and put $A = [A_1 | \cdots | A_K] \in \mathbb{R}^{I \times JK}$, $B = [B_1 | \cdots | B_K] \in \mathbb{R}^{J \times JK}$, $D_k = \text{diag}(0, \dots, 0, 1_J, 0, \dots, 0)$ with the ones in the kth block. Then $X_k = AD_kB^T$ for all k, which is a decomposition with JK terms.

Proof. Part 2. The lower bound I is again from the flattening $[X_1 | \cdots | X_K]$ having rank I almost surely (since $I < JK$). For the upper bound, we need the following.
Claim. If $\text{rank}(\mathcal{X}) = I$ and $JK > I$, then almost surely there is a nonsingular $S \in \mathbb{R}^{I \times I}$ such that $SX_1, \dots, SX_K$ all have proportional rows. Indeed, if $X_k = AD_kB^T$ with $A \in \mathbb{R}^{I \times I}$, then A must be invertible almost surely (otherwise the rows of $[X_1 | \cdots | X_K]$ would not span $\mathbb{R}^I$), and $S = A^{-1}$ gives $SX_k = D_kB^T$, whose ith row is the ith row of B^T scaled by $(D_k)_{ii}$. Conversely, such an S gives a rank I decomposition $X_k = S^{-1}D_k(SX_1)$.
So we need to construct S. Take the weights $d_{ik} = i^{k-1}$ (the Vandermonde matrix), and ask that each row s_i of S satisfies

$$s_i\big(i\,X_1 - X_2\big) = s_i\big(i^2X_1 - X_3\big) = \cdots = s_i\big(i^{K-1}X_1 - X_K\big) = 0 \tag{70}$$

That is, s_i is orthogonal to the columns of the $I \times (JK - J)$ matrix $W_i = [iX_1 - X_2 \,|\, i^2X_1 - X_3 \,|\, \cdots \,|\, i^{K-1}X_1 - X_K]$. Since $I > JK - J$, such a nonzero s_i always exists, e.g. $s_i = H_ie_i$ where $H_i = I - W_i(W_i^TW_i)^+W_i^T$ is the projection onto the orthogonal complement of the columns of W_i. It remains to show that the resulting matrix S is nonsingular almost surely. Its determinant is an analytic function of the entries of $\mathcal{X}$, so it is either identically zero or nonzero almost everywhere (Fisher's theorem). It is not identically zero since there are tensors for which the proportionality condition already holds with $S = I$, and for those the construction returns $S = I$. Hence $\text{rank}(\mathcal{X}) \leq I$ almost surely.

Proof. Part 3. Rank less than I would make the I rows of $[X_1 | \cdots | X_K]$ dependent, which has probability zero. Adding a random horizontal slice gives a $(JK - J + 1) \times J \times K$ tensor which is tall, so has rank $I + 1$ almost surely by (2), and removing a slice cannot increase the rank, so $\text{rank}(\mathcal{X}) \leq I + 1$.
To see that rank I occurs with positive probability, transform the first I columns of $[X_1 | \cdots | X_K]$ to I_I by a change of basis in mode 1, so that the last frontal slice becomes $[Y_{1K}; \dots; Y_{K-1,K}]$ stacked from $K - 1$ blocks of size $J \times J$, while the other slices are block selectors. If all of the Y_{jK} have real eigendecompositions $Y_{jK} = K_j\Lambda_jK_j^{-1}$ (an open condition), then they can be simultaneously diagonalized in the corresponding rows, which gives an explicit decomposition with $I = (K - 1)J$ terms.
Now suppose $K > 2$ and J is odd. Then even without real eigendecompositions we can construct S as before. For each i we need b_i, $d_{i1}, \dots, d_{iK}$ with

$$b_i\Big[i\,Y_{1K} + i^2Y_{2K} + \cdots + i^{K-1}Y_{K-1,K} - d_{iK}I_J\Big] = 0 \tag{71}$$

which says that d_{iK} is an eigenvalue and b_i a left eigenvector of the $J \times J$ matrix $\sum_k i^kY_{kK}$. Since J

is odd, every real $J \times J$ matrix has a real eigenvalue, so we can solve this for each i separately, and nonsingularity of the resulting S holds almost surely by the same analyticity argument. Hence the typical rank is I.

The hypotheses in (3) are sharp. For $K = 2$ the weights i^{k-1} do not differ across blocks, so all the weighted combinations have the same eigenvectors and the argument breaks down, and indeed the typical ranks are $\{I, I+1\}$ by Theorem 6.3. For J even it can happen that every linear combination of the Y_{jK} has only complex eigenvalues, and then no real b_i exists. For instance, for $8 \times 4 \times 3$ tensors, such arrays are rare but arise with positive probability, and both ranks 8 and 9 are typical. Ten Berge's table of results for $K = 2, 3$ is the following.

$K=2$	$J=2$	$J=3$	$J=4$	$K=3$	$J=3$	$J=4$	$J=5$
$I=2$	$\{2,3\}$	3	4	$I=5$	?	?	?
$I=3$	3	$\{3,4\}$	4	$I=6$	6	?	?
$I=4$	4	4	$\{4,5\}$	$I=7$	7	?	?
$I=5$	4	5	5	$I=8$	8	$\{8,9\}$	?
$I=6$	4	6	6	$I=9$	9	9	?

(Entries with $I < J$ follow from the case $I \geq J$ by permuting the modes.) The compact tensors with $I < JK - J$ are not covered by this method, and their typical ranks are only known from numerical computation, which we turn to next.

6.5 Generic Ranks by Computing Dimensions

For the compact cases, the rank can be determined numerically, and the idea is the following. Consider the map

$$\varphi : \{a_r, b_r, c_r\}_{r=1}^{R} \mapsto \sum_{r=1}^{R} a_r \circ b_r \circ c_r \tag{72}$$

from a space of dimension $M = R(I + J + K - 2)$ (we subtract 2 per term since scalings can be moved between the three factors) to $\mathbb{R}^{I \times J \times K}$.

Theorem 6.6 (Comon, Ten Berge, De Lathauwer, Castaing, 2009)

Let $D(R)$ be the rank of the Jacobian of φ at a random point. Then $D(R)$ is the dimension of the closure of the set of tensors of rank at most R. If $D(R) = IJK$, then R is a typical rank, and the smallest such R is the smallest typical rank (the generic rank over $\mathbb{C}$).

Proof. The set of tensors of rank at most R is the image of an irreducible variety under φ, so its closure is irreducible, and the dimension of an irreducible variety equals the dimension of its tangent space at a generic point. By Terracini's lemma, this tangent space is the image of the differential of φ, i.e. the column space of the Jacobian. If its rank is IJK, then the image contains an open set, which has positive volume. Increasing R one at a time until the rank stops increasing gives the smallest R for which the Jacobian has full rank. Explicitly, the rows of the Jacobian for the term r are $I_I \otimes b_r \otimes c_r$, $a_r \otimes I_J \otimes c_r$, $a_r \otimes b_r \otimes I_K$.

Running this for random parameters gives the following smallest typical ranks for cubical tensors of order 3, which are the generic ranks over $\mathbb{C}$.

N	2	3	4	5	6	7	8	9
$N \times N \times N$	2	5	7	10	14	19	24	30

The naive dimension count gives the generic rank as at least $\lceil N^3/(3N-2) \rceil$, which equals 4 for $N = 3$ and not 5, so $3 \times 3 \times 3$ is *defective*. The numerical results also include the tables for 2-, 3-, and 4-slice tensors,

for example the entries $\{2,3\}$ for $2\times2\times2$, $\{3,4\}$ for $3\times3\times2$, $\{5,6\}$ for $5\times3\times3$, which agree with all of the known results above. Moreover, the generic rank is a necessary condition for uniqueness: a tensor of rank larger than the generic value has infinitely many decompositions, whereas a tensor of smaller rank has almost surely finitely many.

6.6 Tensors with Symmetric Slices

When fitting INDSCAL, the tensor has frontal slices that are symmetric, $\mathcal{X}\in\mathbb{R}^{I\times J\times J}$ with each $J\times J$ slice symmetric, and the factors are $A=B$. Symmetry often lowers the typical rank, and it is best understood by looking at what the slice-span lemma says.

Lemma 6.7 (Symmetric Slices)

Let $\mathcal{X}$ have K symmetric $J\times J$ frontal slices. Then $\operatorname{rank}(\mathcal{X})\le J(J+1)/2$. Moreover, if $K\ge J(J+1)/2$ then the typical rank is exactly $J(J+1)/2$.

Proof. The space of symmetric $J\times J$ matrices has dimension $J(J+1)/2$ and is spanned by the rank-one matrices $e_ie_i^T$ and $(e_i+e_j)(e_i+e_j)^T$. The slices lie in this span, so by Lemma 3.1, $\operatorname{rank}(\mathcal{X})\le J(J+1)/2$. For the lower bound, when $K\ge J(J+1)/2$ generic symmetric slices span the whole space of symmetric matrices, so the flattening $X_{(3)}$ has rank $J(J+1)/2$.

This explains the entries in the review where the symmetric typical rank is lower than the non-symmetric one: for $I\times2\times2$ with $I\ge4$, the symmetric typical rank is 3 instead of 4, and for $I\times3\times3$ with $I\ge6$ it is 6 rather than $\min\{I,9\}$. In particular, a $7\times3\times3$ array of symmetric slices is tall, but its typical rank is 6 and not 7, which means that the tall tensor theorem above does not apply to the symmetric case (symmetry does not arise with random sampling). When the slices are not too many, the typical ranks have been determined by Ten Berge, Sidiropoulos, and Rocci for arrays of order $I\times2\times2$ and $I\times3\times3$, and computed numerically by the Jacobian method of Comon et al. for others. For example, for $I\times I\times2$ with symmetric slices the typical rank is again $\{I,I+1\}$, and for $3\times3\times3$ with symmetric slices it is 4, compared to 5 without symmetry.

Lemma 6.8 (Double Centering)

The typical rank of $N_1\times N_2\times N_3$ tensors with all slices **double centered** (every row and column of every $N_2\times N_3$ slice sums to zero) equals the typical rank of free $N_1\times(N_2-1)\times(N_3-1)$ tensors.

Proof. A double centered slice lies in $\mathbf{1}^\perp\otimes\mathbf{1}^\perp$, and a basis of $\mathbf{1}^\perp$ identifies this with $\mathbb{R}^{N_2-1}\otimes\mathbb{R}^{N_3-1}$. Applying the projection onto $\mathbf{1}^\perp$ in modes 2 and 3 fixes the tensor and maps rank-one terms to rank-one terms, so the rank can be computed using only centered factors. Therefore the rank is the same as that of the corresponding $N_1\times(N_2-1)\times(N_3-1)$ tensor, and the change of basis is a linear isomorphism, which does not change the rank.

Finally, for the fully symmetric case, the numerical method recovers the Alexander–Hirschowitz values: the generic symmetric rank of cubic forms in $N=2,\dots,8$ variables is $2,4,5,8,10,12,15$, with the exceptional value 8 for $N=5$ (compared to the formula value 7), and for quartics in $N=2,\dots,8$ variables it is $3,6,10,15,21,30,42$.